

Quasi-experimental evaluation of a nationwide diabetes prevention programme

<https://doi.org/10.1038/s41586-023-06756-4>

Received: 19 May 2022

Accepted: 17 October 2023

Published online: 15 November 2023



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Diabetes is a leading cause of morbidity, mortality and cost of illness^{1,2}. Health behaviours, particularly those related to nutrition and physical activity, play a key role in the development of type 2 diabetes mellitus³. Whereas behaviour change programmes (also known as lifestyle interventions or similar) have been found efficacious in controlled clinical trials^{4,5}, there remains controversy about whether targeting health behaviours at the individual level is an effective preventive strategy for type 2 diabetes mellitus⁶ and doubt among clinicians that lifestyle advice and counselling provided in the routine health system can achieve improvements in health^{7–9}. Here we show that being referred to the largest behaviour change programme for prediabetes globally (the English Diabetes Prevention Programme) is effective in improving key cardiovascular risk factors, including glycated haemoglobin (HbA1c), excess body weight and serum lipid levels. We do so by using a regression discontinuity design¹⁰, which uses the eligibility threshold in HbA1c for referral to the behaviour change programme, in electronic health data from about one-fifth of all primary care practices in England. We confirm our main finding, the improvement of HbA1c, using two other quasi-experimental approaches: difference-in-differences analysis exploiting the phased roll-out of the programme and instrumental variable estimation exploiting regional variation in programme coverage. This analysis provides causal, rather than associational, evidence that lifestyle advice and counselling implemented at scale in a national health system can achieve important health improvements.

Diabetes prevalence and diabetes-related deaths continue to rise in most parts of the world^{1,11}. By 2030, the number of adults with diabetes globally is expected to reach 578 million, representing 10% of the global adult population¹¹. Consequently, there is an urgent need to implement population-based measures that prevent diabetes, enhance its early detection and address cardiovascular risk factors to prevent or delay its progression to complications.

In the seminal US Diabetes Prevention Program study (which serves as a model for many behaviour change programmes in the USA and elsewhere)¹², targeting changes in individuals' lifestyle behaviour was more successful than administering metformin in preventing or delaying diabetes. However, clinical trials such as the US Diabetes Prevention study have mainly focused on efficacy, supplying proof of principle that the intervention worked when extensive one-to-one sessions with specialists and a range of incentives are being provided¹³. Thus, although a recent meta-analysis concluded that lifestyle modification provides strong evidence for reversing prediabetes in adults⁵, it remains important to establish the transferability of behaviour change programmes

into real-world settings, in which resources and support are generally more limited than in the clinical research setting.

Establishing that behaviour change programmes work in routine care is essential for several reasons. First, although lifestyle counselling is the recommended first-line treatment option for people presenting with prediabetes and other cardiovascular risk factors in clinical guidelines¹⁴, there is ample evidence from a variety of settings that clinicians often do not follow these guidelines^{15–20} and instead revert to prescribing preventive medication because of limited time resources in primary care²¹, insufficient knowledge and referral options for promoting healthy lifestyles^{7,22} and a predominance of the biomedical model with clinicians being uncertain about the success of counselling^{7,8}. In particular, doubt that behaviour change at the level required for substantial weight loss is possible to achieve for most patients is prevalent among primary care clinicians⁹. Second, participants in clinical trials for behaviour change programmes are unlikely to be representative of the broader patient population. For instance, patients enrolled in clinical cardiology trials had a lower risk profile as they tended to be

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younger, male and less likely to have a comorbid disease than individuals encountered in everyday practice²³. A general population sample may also respond differently to behaviour change programmes because of lower health literacy and willingness to engage, higher comorbidities and greater ethnic diversity^{24,25}.

Using routine health data

We advance the argument that the impact of behaviour change programmes on population health must relate to real-world effectiveness and should, thus, be evaluated in an “observational, non-interventional trial in a naturalistic setting” akin to phase 4 in drug development^{26,27}. However, conventional observational studies that are generally applied in health research have the disadvantage that they may fail to account for principal confounding factors such as selection biases and, thus, preclude causal interpretations²⁸. In contrast, we establish causality by applying a regression discontinuity approach, one of the most credible quasi-experimental strategies for causal inference^{10,29}, that combines large-scale routine data from the English National Health Service (NHS) with variation in treatment probabilities generated by guidelines from the National Institute for Health and Clinical Excellence (NICE) that recommend intensive lifestyle counselling for people at high risk of progression to type 2 diabetes mellitus (T2DM)¹⁴. Specifically, we use the fact that the NHS Diabetes Prevention Programme (NHS DPP), a behaviour change programme with weight loss, diet and physical activity goals consisting of at least 13 group sessions over the course of 9 months and the largest DPP globally to achieve universal population coverage, is only open to patients above a prespecified threshold of HbA1c or fasting plasma glucose indicating non-diabetic hyperglycaemia³⁰. By exploiting this eligibility threshold that draws a sudden distinction between the prediabetes and normal glycaemia range, we can take advantage of existing large-scale routine health data while still obtaining causal effect estimates that are not vulnerable to confounding and measurement error (Supplementary Information section 1.1 gives an overview of existing correlational evidence for the NHS DPP)^{10,29}.

Our study used data from the Clinical Practice Research Datalink (CPRD) Aurum and NHS England’s Hospital Episode Statistics (HES). CPRD Aurum is a large primary care database of de-identified electronic health records from a network of about one-fifth of general practitioner (GP) practices across England. To ensure sufficient implementation of the NHS DPP during the study period after the start of the phased roll-out in mid-2016, our population of interest consisted of adults (aged 18 to 80 years) who received an HbA1c test between 1 January 2017 and 31 December 2018. Data were available until the end of June 2020. We identified 2,106,376 patients who had a baseline HbA1c test during the enrolment and met inclusion criteria (Methods section, ‘Data source and study population’). Patient characteristics are described in Extended Data Table 1.

In all our analyses, the primary outcome was change in HbA1c. For the regression discontinuity analysis, secondary outcomes included changes in body mass index (BMI), body weight, blood pressure, serum cholesterol concentrations and serum triglyceride concentrations. We also conducted exploratory analyses investigating the effect of programme referral on the probability of diabetes, hypertension and hyperlipidaemia incidence; receipt of newly prescribed medications for these conditions; diabetes complications; all-cause mortality; and emergency hospitalization for a major adverse cardiovascular event (MACE). A detailed definition of each outcome is provided in Supplementary Information section 1.2.

Assumptions for causal effect estimation

In the first part of the analysis, we ensured that all necessary assumptions for a regression discontinuity analysis were met³¹. In particular, we

assessed the continuity assumption²⁹, which requires that the density distribution of baseline HbA1c be continuous around the prediabetes threshold of 42 mmol mol⁻¹ (6%). This assumption would be violated if patients or providers could precisely manipulate baseline HbA1c, which is virtually impossible in our scenario given that the exact blood test result is automatically uploaded into the electronic health data system. We plotted the density distribution of baseline HbA1c values around the threshold and, as expected, found no evidence of heaping or manipulation (Supplementary Fig. 1).

Receipt of treatment was defined as a record of a referral to a behaviour change programme or intensive lifestyle counselling during the 12 months after the baseline HbA1c test. Treatment primarily included referrals to the NHS DPP but we also included referrals to other structured programmes and intensive lifestyle counselling as they are likely to serve as an alternative when placement in NHS DPP is not possible. We conducted analyses testing the sensitivity of results to this choice (Methods section, ‘Outcome and treatment definition’). For convenience only, we henceforth refer to these treatments simply as intensive lifestyle counselling. We observed a 10.8 percentage points increase in treatment assignment at the prediabetes threshold (Fig. 1a). In relative terms, patients just above the threshold were five times more likely to be referred to intensive lifestyle counselling compared to patients just below the threshold. There is no pathophysiological phenomenon at that specific threshold value; rather, increasing HbA1c is associated with an increased risk of developing T2DM and cardiovascular disease in a more continuous fashion³². The somewhat arbitrary nature of the prediabetes HbA1c threshold along with the fact that HbA1c is measured with a degree of random measurement error means that patients just below and above the threshold should be close to identical in their underlying characteristics. In Fig. 1b–d, we show that, unlike for referral to intensive lifestyle counselling, there is no sudden ‘jump’ at the prediabetes HbA1c threshold in age, baseline BMI and frequency of GP visits, signifying that these variables cannot confound our analysis. We demonstrate this further using balance tests, empirically showing that a comprehensive set of potential confounding variables, including demographic characteristics, physiological variables and markers of healthcare use and literacy, were all continuously distributed near the threshold (Supplementary Figs. 2 and 3 and Extended Data Table 2). Taken together, our initial analyses provide strong evidence that the necessary assumptions for a valid regression discontinuity analysis were met¹⁰, meaning that patients below and above the threshold were effectively randomized to being referred to the programme or not, as they would be in an ideal target trial (Supplementary Information section 1.3).

Establishing whether glycaemic control improves

Next, we proceed to compare patients lying closely on either side of the threshold using local linear regression. We found that patients who were referred to intensive lifestyle counselling significantly improved their HbA1c concentrations. Specifically, we evaluated the effect of referral to intensive lifestyle counselling on glycaemic control by fitting separate regression lines of the association between baseline HbA1c and change in HbA1c above and below the eligibility threshold. The difference in where these lines intersect at the threshold quantifies the discontinuity in the outcome and can be described as the effect of the threshold rule (−0.10 mmol mol⁻¹, 95% CI −0.16, −0.03). This effect merely measures the effect of being eligible for intensive lifestyle counselling as determined by the clinical guideline rather than the effect of being referred to intensive lifestyle counselling. Given that only 17.4% of eligible patients were referred to intensive lifestyle counselling, it is necessary to scale the effect of the threshold rule by the difference in the probability of treatment at the threshold to determine the effect of being referred to intensive lifestyle counselling. When doing this, we find a significant beneficial effect of referral to intensive lifestyle

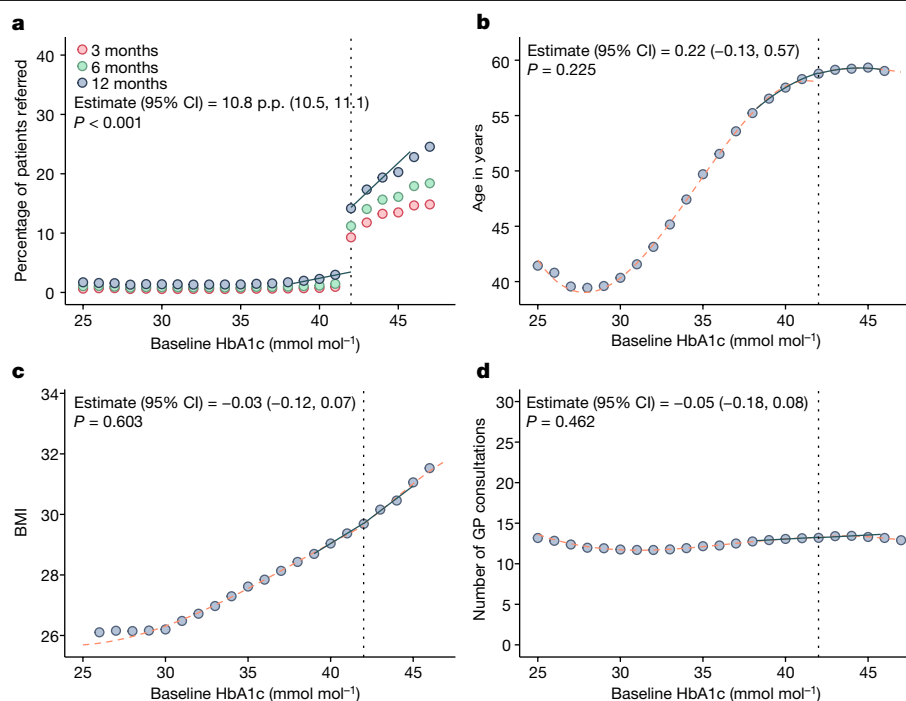


Fig. 1 | Association between baseline HbA1c and intensive lifestyle counselling and potential confounders. a, Primary exposure ('first stage'), that is, referral to intensive lifestyle counselling at 3, 6 and 12 months after baseline HbA1c. **b–d**, Potential confounders: age (**b**), baseline BMI (**c**) and prior GP consultations (**d**). The blue lines show the local linear regression models in the bandwidth used in our primary analysis (Methods, section 'Main analysis'). For age (**b**), we assumed a quadratic relationship with baseline HbA1c. The orange

dotted lines show the global polynomial relationship. The blue circles represent the mean value for individual patients and the dotted vertical lines indicate the HbA1c cutoff. The estimate represents the discontinuity at the HbA1c threshold, whereas discontinuities in potential confounders may jeopardize assumptions underlying regression discontinuity. Significance was tested in two-sided *t*-tests (*P* < 0.05). The sample size in the bandwidth is *n* = 513,473 (**a**, **b** and **d**) and *n* = 332,156 (**c**). p.p., percentage points.

counselling on HbA1c at follow-up ($-0.85 \text{ mmol mol}^{-1}$, 95% CI -1.46 , -0.24). Details can be found in the Methods.

Although the clinical significance of a $0.85 \text{ mmol mol}^{-1}$ reduction in HbA1c is difficult to quantify at an individual level, observational clinical data suggest a linear association between HbA1c and cardiovascular disease, even in non- or prediabetic individuals³². For example, as a reference point, after adjusting for principal conventional cardiovascular risk factors, individuals having HbA1c concentrations of 5.5–5.7% (which approximately translates to 37–39 mmol mol^{-1}) were almost twice as likely to be diagnosed with coronary artery disease compared to individuals with less than 5.5% (ref. 33). Thus, it is likely that a reduction of $0.85 \text{ mmol mol}^{-1}$ is meaningful at the population level.

The effects of the threshold rule (of being eligible for a referral) and of being referred to intensive lifestyle counselling for all outcomes are shown alongside optimal bandwidth and sample size in Extended Data Table 3. To ensure that the finding of improved glycaemic control is not sensitive to our selected bandwidth or functional form (linear or quadratic), we show that effect sizes were robust to different choices (Fig. 2a, Supplementary Fig. 4 and Supplementary Tables 2 and 3). We also performed the regression discontinuity analysis using a secondary cohort, for which we only included patients whose baseline HbA1c test was their first ever recorded HbA1c test. Here, being referred to intensive lifestyle counselling led to a $1.26 \text{ mmol mol}^{-1}$ reduction in HbA1c (95% CI -2.06 , -0.46). All results for the secondary cohort can be found in Supplementary Information section 3.

Intent-to-treat versus per-protocol effect

Our estimated effects are relevant for clinicians and policy-makers who need to understand the effects of deciding to refer patients to intensive lifestyle counselling (the 'intent-to-treat' effect), regardless of

whether patients then decide to attend the programme, which is largely not in the clinician's control. However, understanding programme efficacy (the 'per-protocol' effect or, in other words, the effect of the programme if everyone who is referred attends) also provides useful information. It is important to consider that programme attendance may be a function of individual characteristics such as personal motivation or available time resources. We assessed the amount of adherence by determining whether the patient started the intervention sessions of the NHS DPP according to GP records. In line with statistics from an official programme evaluation³⁰, among eligible patients that were referred to the NHS DPP in our primary cohort, 28.1% of patients started the intervention. By scaling up the effect of programme referral in the selected bandwidth, we approximate that patients' HbA1c concentration would be reduced by $3.0 \text{ mmol mol}^{-1}$ if patients perfectly adhered to referrals and took up the programme. Patients who started the treatment are largely similar to patients who were referred but did not start the treatment, in particular on variables that may indicate heightened personal motivation or health literacy such as attending preventative cancer screening services or vaccine uptake (Supplementary Table 4) and our main results were robust to adjusting for variables that may serve as a proxy for personal motivation and health literacy (Supplementary Table 5).

Ruling out confounding and bias

It is important to rule out that we may falsely attribute beneficial effects to being referred to intensive lifestyle counselling whereby they were in fact induced by medication. When adjusting our results for being prescribed diabetes medication during follow-up, the estimated causal effect of referral to intensive lifestyle counselling on glycaemic control indicated a larger reduction, suggesting that improvements in

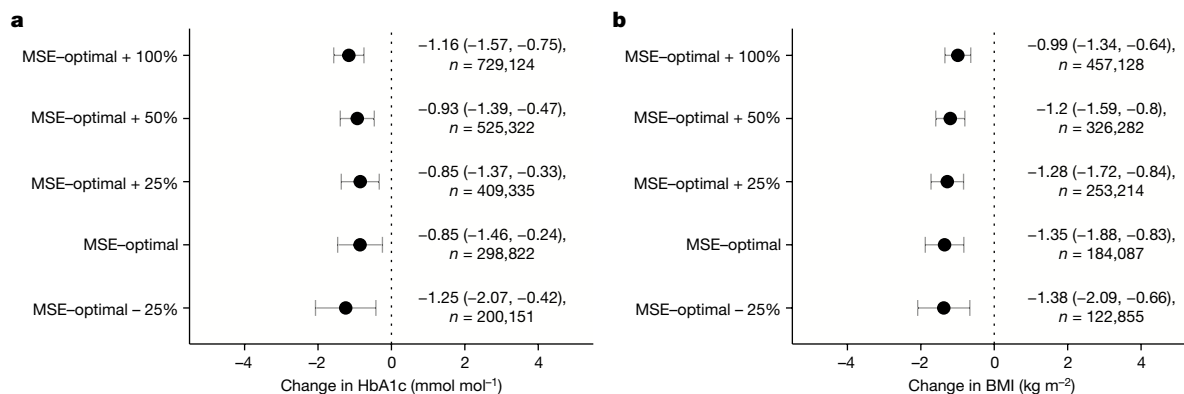


Fig. 2 | Robustness of the effects of being referred to intensive lifestyle counselling on HbA1c and BMI across bandwidth choices. The mean-squared error (MSE)-optimal bandwidth is 3.8 mmol mol⁻¹ below and above the threshold. **a,b**, The estimated complier average causal effect of referral to intensive lifestyle counselling on change in HbA1c (**a**) and change in BMI (**b**) from local linear

regressions with varying bandwidths (75%, 125%, 150% or 200%) of the MSE-optimal bandwidth with heteroskedasticity-robust 95% CI and triangular kernel weights. The sample size of patients in each bandwidth is given alongside the effect estimates. All effect estimates are statistically significant in two-sided *t*-tests ($P < 0.05$).

HbA1c were, indeed, not driven by increased use of diabetes medication (Extended Data Table 3). In general, having an HbA1c concentration above the eligibility threshold for the NHS DPP was associated with a small increase in the probability of being prescribed diabetes medication shortly after treatment assignment (risk difference in percentage points (RD) = 0.04, 95% CI 0, 0.09), which increased to 0.3 percentage points at follow-up. However, the discontinuity in the probability of being prescribed diabetes medication was not significant when using robust bias-corrected confidence intervals for inference (Extended Data Table 4). Specifically, out of 26,513 patients with a baseline HbA1c between 42 and 47 mmol mol⁻¹ who were referred to intensive lifestyle counselling, only 882 (3.3%) were prescribed diabetes medication during the 12 months following treatment assignment with numbers increasing with increasing HbA1c concentrations; these numbers are unlikely to substantially impact improvements in glycaemic control. There was no discontinuity in newly prescribed lipid-lowering medication (RD = 0.29, 95% CI -0.23, 0.82) or blood pressure-lowering medication (RD = 0.11, 95% CI -0.38, 0.60).

We further performed a set of robustness checks to detect uncontrolled confounding or other sources of bias following the approach by ref. 34. First, we did a placebo analysis using an alternative patient cohort that is comparable in inclusion criteria to our primary analysis but who had baseline HbA1c measures in 2014 and 2015. During this period, patients were already identified with prediabetes on the basis of the same threshold and public health guidance. However, the key difference was that there were limited intensive lifestyle programmes in place to which GPs could refer their patients. This led to a scenario in which key potential confounders such as discontinuities in outcome availability and number of consultations could already be observed (Supplementary Fig. 5), whereas the proposed causal mechanism—the NHS DPP—was ‘deactivated’ (that is, a negative exposure control scenario; Supplementary Fig. 6a). We find that being eligible for the NHS DPP in these years in which the NHS DPP was not yet available did not result in an improvement in HbA1c or BMI (Supplementary Fig. 6b,c). Second, we repeated the main regression discontinuity analysis for health outcomes which we did not expect to be plausibly affected by referral to the NHS DPP (so-called negative outcome controls). Results indicate no systematic confounding in our primary cohort (Extended Data Table 5). Third, our results are robust to adjusting for the number of months between baseline and endline measurement (Supplementary Tables 2 and 3). Fourth, we restrict our sample to patients with a follow-up period of at least 12 months (and 18 months, respectively). Although some of the observed effect estimates are smaller and show greater

uncertainty compared to the full sample, they still point towards health gains comparable to those in our primary analysis (Supplementary Tables 6 and 7).

Improvements in other health outcomes

In secondary analyses, we found evidence that other key cardiovascular risk factors improved. Referral to intensive lifestyle counselling significantly reduced BMI (-1.35 kg m⁻², 95% CI -1.88, -0.83; Fig. 2b) and weight (-2.99 kg, 95% CI -4.38, -1.61). Albeit not significant, effect estimates were also in the direction of benefit for blood pressure levels (diastolic: -1.35 mmHg, 95% CI -3.31, 0.61; systolic: -2.03 mmHg, 95% CI -4.96, 0.91). When adjusting results for the prescription of blood pressure-lowering medication, improvements in systolic blood pressure persisted and became marginally significant ($P = 0.092$; Extended Data Table 3). Referral to intensive lifestyle counselling also significantly reduced triglyceride concentrations (-0.33 mmol l⁻¹, 95% CI -0.54, -0.12) and increased HDL concentrations (0.04 mmol l⁻¹, 95% CI 0, 0.09). There was no significant effect on other serum cholesterol concentrations (LDL and the total cholesterol-to-HDL ratio) and no effect on the probability of being prescribed lipid-lowering medication. Results were robust to adjusting for baseline observables (Supplementary Table 2) and using different bandwidths (Supplementary Figs. 7–10) or a quadratic model (Supplementary Table 3). Results for our secondary cohort yielded a very similar pattern of results but slightly larger effect sizes (Supplementary Information section 3).

Diabetes complications, emergency hospitalization for MACEs and mortality were not significantly reduced by being referred to intensive lifestyle counselling in exploratory analyses (Extended Data Table 3). The low incidence of adverse downstream events during our relatively short follow-up period, with 26,567 (1.3%) of patients dying and 36,567 (1.8% of those 2,037 384 linkable to HES data) having an emergency hospitalization for a MACE, resulted in low statistical power for detecting small, short-term changes in these outcomes. We observed a significant increase in T2DM diagnoses at the threshold (RD = 0.39, 95% CI 0.21, 0.57; Extended Data Table 3). We interpreted these results cautiously because diagnoses in electronic health records may be less reliable than biochemical measures³⁵ and a stronger focus on identifying and monitoring people who are at risk of T2DM is likely to initially increase the incidence of T2DM independent of health improvements. Hypertension and hyperlipidaemia incidence did not significantly change at the threshold (Supplementary Tables 2 and 3).

Men may be benefitting more than women

Results stratified by gender, age group, ethnicity, socioeconomic status (based on the Index of Multiple Deprivation (IMD) which is derived from the patient's postcode) and rural or urban practice location are presented in Supplementary Tables 8–16. Stratification led to relatively small sample sizes for practices in rural locations and patients with Asian, Black or mixed ethnicity and in the youngest age group. Being referred to an intensive lifestyle intervention led to significant improvements in HbA1c, weight, blood pressure and triglycerides in men but not women (Supplementary Tables 8, 10–12 and 16). Both men and women significantly improved their BMI although effect estimates suggest larger improvements in men compared to women (Supplementary Table 9). There was no indication that a higher socioeconomic status was consistently associated with greater benefits. We further stratified results by variables that may function as a proxy for health literacy, health services use and personal motivation (Supplementary Tables 8–16). Although the results are exploratory and limited by the absence of prespecified hypotheses and insufficient statistical power, effect estimates suggest that patients who did not receive lifestyle advice from their GP previously were more likely to benefit from being referred to intensive lifestyle counselling.

Triangulation of evidence

To further boost confidence in our findings, we triangulate the evidence from our regression discontinuity approach by (1) performing a difference-in-differences analysis, an established quasi-experimental method originating from social sciences that leverages the phased roll-out of the NHS DPP in three waves and (2) using the regional variation in programme coverage as an instrumental variable for programme referral (Methods)³¹.

In the difference-in-differences analysis our estimate of interest represents how much the average outcome of the treatment group has changed in the period after the treatment, compared to what would have happened in the absence of the intervention. A key assumption is that, in the absence of treatment, the outcome trends would have been the same in the treatment group as in the comparison group³¹. We interrogated this assumption by examining whether the trends in HbA1c are visually parallel in the pre-introduction period (Extended Data Fig. 1) and by testing the interaction between wave and (pre-introduction) year in linear regressions (Wald test: wave 1, $P = 0.273$, wave 2, $P = 0.809$). The group–time average treatment effect estimates provide support for the view that implementing the NHS DPP led to an improvement in glycaemic control (Extended Data Table 6a). For each relevant comparison during the post-DPP introduction, there is a statistically significant beneficial effect on HbA1c (wave 1 versus wave 3 in 2018/2019: $-0.126 \text{ mmol mol}^{-1}$, 95% CI -0.235 , -0.018 ; wave 1 versus wave 3 in 2019/2020: $-0.328 \text{ mmol mol}^{-1}$, 95% CI -0.447 , -0.209 ; and wave 2 versus wave 3 in 2019/2020: $-0.140 \text{ mmol mol}^{-1}$, 95% CI -0.231 , -0.048 ; Fig. 3a). When we consider how the effect of implementing the NHS DPP changes by the length of time that the programme has been in place, these parameters paint largely the same picture as the group–time average treatment effects (Fig. 3b). The effect of implementing the NHS DPP on HbA1c is negative and increasing in magnitude the longer practices have implemented the NHS DPP. By applying a conditional parallel trends assumption in a secondary analysis, we ensured that the results were not confounded by the extension of routine HbA1c testing during programme roll-out (Methods section, 'Difference-in-differences analysis'; Extended Data Table 6b and Supplementary Fig. 11).

As the NHS DPP was not rolled out in parallel across regions in England (Extended Data Fig. 2), we were able to leverage the time-variant regional differences in NHS DPP coverage as an instrument for actual referral to the programme. Specifically, we estimated the effect of NHS

DPP referral on glycaemic control from comparing those who changed from not being referred to being referred because of a change in programme coverage in their region. Although this instrumental variable analysis is relatively imprecise because it relies on time-by-region variation instead of patient-level variation, we continue to find a significant beneficial effect of programme referral on endline HbA1c (result from two-stage least squares (2SLS) regression with practice- and year-fixed effects: $-3.77 \text{ mmol mol}^{-1}$, 95% CI -7.52 , -0.01).

Finally, we also provide correlational evidence from propensity score matching, multivariate regression and panel regression with individual- and year-fixed effects (Methods section, 'Multivariate regression analyses, propensity score matching and panel regression'). The results of these analyses were consistent with those from our regression discontinuity approach (Supplementary Table 17). We provide an overview of all effect estimates measuring the change in HbA1c in Extended Data Table 7.

Discussion

Although clinical trials have shown that intensive lifestyle counselling is efficacious in improving cardiovascular risk factors in controlled research settings^{4,5,36–38}, we present evidence that these health benefits can be successfully translated to and scaled up in routine care. In our study using electronic health records from more than 2 million patients, we found causal evidence that referral to the largest behaviour change programme for prediabetes globally led to improved glycaemic control and reductions in BMI, weight, HDL and triglycerides. Systematic reviews and meta-analyses of controlled clinical trials studying effects of lifestyle interventions (including diabetes prevention programmes in US settings) on weight loss and blood pressure in adults with prediabetes found effect sizes comparable to those in our study^{36,38–40}. Although evidence on improvements in glycaemic control from controlled trials seems more mixed^{38–40}, our study substantiates indications from earlier correlational studies that suggest beneficial effects of NHS DPP participation on HbA1c and weight (Supplementary Information section 1.1). We discuss results from secondary and stratified analyses in the Supplementary Discussion.

We demonstrated the use of a regression discontinuity approach for evaluating a population-wide health service intervention by leveraging a threshold in treatment assignment induced by clinical guidelines. Thresholds are ubiquitous in clinical medicine and, thus, represent a rich opportunity to generate causal, rather than associational, evidence of treatment effectiveness. In conjunction with increasing access to routine electronic health records and detailed health information, regression discontinuity analyses have the potential to advance evidence-based healthcare and implementation science. For example, many conceivable research questions that are of interest to clinical medicine and health systems research cannot be studied in conventional or pragmatic randomized trials, either for feasibility (such as very long follow-up periods to establish the effectiveness of anti-ageing agents) or ethical reasons (such as withholding treatments that have become the standard of care). Regression discontinuity designs may also be used to investigate aspects of health equity, for example, heterogeneous treatment effects between patient groups linked to sociodemographic characteristics and comorbidities, for which randomized controlled trials usually have too small a sample size or an insufficiently diverse study population.

Inherent to any regression discontinuity analysis is the limitation that we can only estimate the causal effect for those who initiate the treatment because they crossed the eligibility threshold. This effect may differ from the (unobserved) treatment effects for patients who would have been referred to intensive lifestyle counselling regardless of baseline HbA1c concentrations (the 'always takers'), for example, because of clinical symptoms or patients who would not have participated in any programme or counselling even if eligible (the 'never

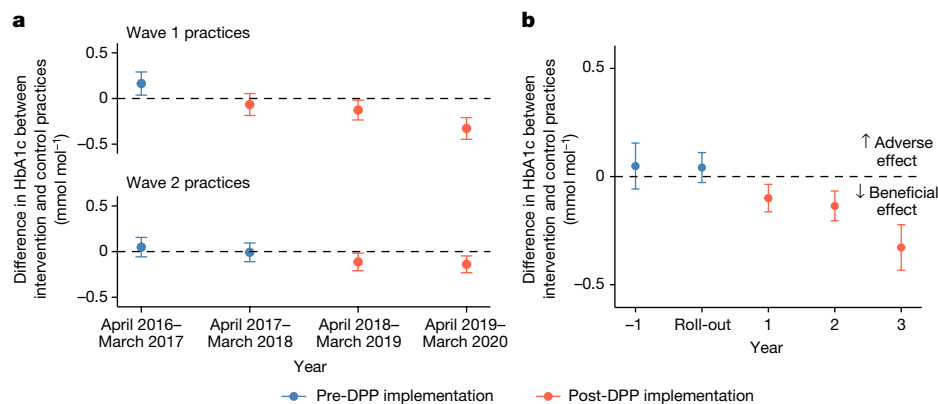


Fig. 3 | Difference-in-differences effect estimates of NHS DPP

implementation. a, Group-time average treatment effects for patients from wave 1 and wave 2 compared with patients from wave 3 practices. **b**, Partially aggregated average effect by length of exposure, that is, programme implementation, in years. Roll-out for wave 1 practices started in June 2016

and roll-out for wave 2 practices started in April 2017. Effect estimates are presented with 95% CI on the basis of standard errors clustered at the practice level using multiplier bootstrap procedures (Methods section, ‘Difference-in-differences analysis’).

takers’). Also, effects may not be generalizable to those further away from the HbA1c threshold that defines prediabetes. This is, however, not a concern in this study given that the NHS DPP specifically targets people with prediabetes, which in turn is defined by a fairly narrow HbA1c range. In our analysis, we mainly relied on the complier average causal effect (CACE), estimating the effect for those who were actually referred to an intervention. Given that there was a large percentage of individuals presenting above the HbA1c threshold who were not referred to any lifestyle counselling, it is important to not mistake the observed CACE effects for the population health effect of the NHS DPP. Although the NHS DPP is operating at a large scale, with 100,000 referrals being offered in 2021 (ref. 41), there remains a substantial proportion of adults in England with impaired glycaemic control who are presently not taking part in intensive lifestyle counselling, whether it is because of system-level (for example, placements in the NHS DPP are not available) or physician- and patient-level reasons (for example, GPs are not compliant with the guidelines). Especially given low uptake and high attrition rates among socio-economically disadvantaged and diverse ethnic groups⁴², our study lends support to calls for further investment in behavioural interventions and targeted prevention strategies for individuals at risk for diabetes that are at present not reached through care pathways. It should be further investigated whether some groups may benefit from combining programmes such as the NHS DPP with continuing, long-term support through social prescribing and community engagement initiatives that increases attention to patients’ wider social context and health choice architecture⁴³.

This study has several other limitations. First and foremost, we observed considerable missingness in outcomes such as HbA1c or BMI, which were also measured more frequently by GPs in the follow-up period if patients had crossed the prediabetes threshold. This differential missingness in our outcome variables will be a source of bias if individuals with missing outcome data have systematically different outcome values compared with those who do not. Having said that, we show that crossing the prediabetes threshold and subsequent closer monitoring of biomarkers by GPs was not associated with improvements in our outcome variables before NHS DPP roll-out. Closer monitoring of patients, however, may be the reason for which we find that referral to intensive lifestyle counselling increased the probability of being diagnosed with T2DM independently of improvements in biomarkers.

Another potential limitation is violation of the exclusion restriction, whereby treatments or exposures other than intensive lifestyle counselling are affected by crossing the cutoff. Although we could

not precisely control for the relationship between drug dosage and secondary outcomes, we are confident that the observed health effects were not attributable to increased medication following treatment assignment as effect estimates were robust to adjusting for drug prescriptions. In addition, testing a diverse set of negative outcome controls suggests no systematic confounding. We further conditioned our results on variables that may function as a proxy for health literacy, health services use and personal motivation, which did not substantially change our results and performed a robustness analysis restricting our sample to patients with regular GP visits before their treatment assignment.

Finally, because of the use of electronic health records we had no detailed information about adherence or persistence to behaviour change programmes and lifestyle counselling over time. Early evidence suggests that the number of attended sessions in the NHS DPP acts as a mediator for successful weight loss at follow-up⁴⁴. However, our primary interest was in whether the benefits of the NHS DPP are seen in real-world circumstances, outside the controlled conditions of a randomized controlled trial. Non-adherence to the programme, for example because patients feel that the visits are too frequent or lengthy, are part and parcel of such a real-world effectiveness assessment. We discuss limitations of our two other quasi-experimental approaches; difference-in-differences analysis and instrumental variable estimation in the Supplementary Discussion.

Taken together, our findings demonstrate that referral for intensive lifestyle counselling in routine care, in the form of a population-wide diabetes prevention programme, is effective in improving glycaemic control and reducing body weight. Scepticism about the effectiveness of lifestyle counselling for successful behaviour change may stem from clinicians’ experience that brief lifestyle counselling—that is often the only feasible approach in time-constrained GP consultations—may be of no or very limited benefit. Investments in structured, intensive behaviour change programmes, on the other hand, may help to promote primary and secondary prevention of T2DM and reduce the risk of complications from diabetes and cardiovascular events. Their positive effects may also extend to other non-communicable diseases such as cancer, which is increasingly thought to be connected to unhealthy lifestyle habits and environments⁴⁵ or communicable diseases such as influenza or COVID-19, which more gravely affect people with known cardiovascular risk factors such as diabetes⁴⁶. Thus, our study not only demonstrates the potential of intensive lifestyle counselling for improving the health of patients with prediabetes in routine care but potentially also suggests a promising route for improving population health more broadly.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-023-06756-4>.

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Description of the NHS Diabetes Prevention Programme

UK NICE public health guideline 38 recommends that people who have non-diabetic hyperglycaemia (HbA1c concentration of 42–47 mmol mol⁻¹ (6.0–6.4%) or fasting plasma glucose of 5.5–6.9 mmol l⁻¹) and are thus at high risk of progression to type 2 diabetes (T2DM) are referred to a “local, evidence-based, quality-assured intensive lifestyle change programme” to prevent or delay the onset of T2DM¹⁴. The NHS DPP began phased roll-out in 2016. The provider contracts require the intervention to be delivered face-to-face to groups of 15–20 adults over at least 13 sessions (totalling 16 h) with a minimum of 9 months’ duration, with the aim of supporting behaviour change to result in improved diet, increased physical activity and weight loss. Activities include a mixture of education, group support, knowledge testing, visual activities and activities led by patients^{47,48}. Eligible individuals (a recorded HbA1c measure of 42–47 mmol mol⁻¹ in the past 12 months or a history of gestational diabetes) are generally identified by their GP for inclusion in the programme. HbA1c concentrations may be measured as part of routine clinical practice or an NHS Health Check (a health check-up offered by GPs for adults in England aged 40 to 74 years to detect early signs of stroke, kidney disease, heart disease, type 2 diabetes or dementia)³⁰. An option to self-refer based on the Diabetes UK ‘Know your Risk’ tool had been added temporarily as a referral route during the Covid-19 pandemic (after our study period)⁴⁹. According to guidelines, people below the HbA1c eligibility threshold for the programme should be offered brief advice or intervention and receive information about services that could help them change their lifestyle, bearing in mind their risk profile. Our analysis studies the original framework of the NHS DPP before changes such as more commercial providers or the provision of a digital option were introduced in 2020 (ref. 50).

Initial projections by Public Health England and NHS England expected that GP identification would generate demand for around 100,000 participants a year once the programme was rolled out nationally by 2020. Official reported statistics from the National Diabetes Audit confirm that referral numbers (offers that were made and not declined) increased steadily from 2017 to 2020, with reporting periods covering January to March of the next year exceeding modelled targets as early as 2018 (2017, 103,295; 2018, 241,255; 2019, 386,025; and 2020, 559,770)⁵¹. In a recent analysis from the DIPLOMA research programme (Diabetes Prevention—Long Term Multimethod Assessment), it was reported that by April 2020, providers had received a total of 513,312 referrals, of which 271,208 (52.8% of total referrals) had attended an initial assessment and 101,175 (19.7% of total referrals) had attended at least 60% of the programme’s sessions⁵⁰. Publications report on uptake rate and progression through the programme in different ways. According to a scoping review published in early 2022, uptake of the initial assessment following a referral varied from 40% to 78% (ref. 42). Others report on attendance at the first intervention session with rates varying from 36% to 94% (ref. 42). Similarly, numbers on how many participants reached the mid- or end-point of the programme vary considerably. According to an early service evaluation, as of the end of December 2018, 36% attended at least one intervention session and 19% attended at least 60% of the sessions³⁰. In this early service evaluation, younger age, Asian and mixed ethnicity (compared to white ethnicity), a lower socioeconomic status and baseline obesity were associated with lower rates of completion.

Data source and study population

The study used data from the CPRD Aurum⁵² and the NHS England Hospital Episode Statistics Admitted Patient Care (HES APC) database⁵³. CPRD Aurum is a large primary care database of de-identified electronic health records from a network of about one-fifth of GP practices across the United Kingdom. The data are representative of the broader English

population in terms of geographical coverage, socioeconomic deprivation, age and gender⁵². In July 2020, anonymized longitudinal data from 35.9 million patients and 1,296 at present contributing English GP practices were available. HES APC is a secondary care database in England and records patient data related to presentations in NHS hospitals or private healthcare institutions for which the NHS provides partial funding. It covers all NHS hospitals in the country. The study protocol was approved by the Independent Scientific Advisory Committee (ISAC) for MHRA research (protocol no. 20_000052). Ethics approval for observational research using the CPRD and linked anonymized NHS healthcare data with approval from ISAC was granted by a National Research Ethics Service committee (Multiple Research Ethics Committee ref. 05/MRE04/87).

In contrast to CPRD Aurum data, which do not contain lifetime follow-up of patients because patients are entering the database when they register with a contributing GP practice and exiting the database when they leave that practice, patients in HES APC maintain the same ID throughout their time in the database⁵⁴. As this information was only available for patients linkable to HES APC, we kept all patients in the analysis regardless of whether they seemed under several CPRD IDs. Results were insensitive to dropping all patients with several CPRD IDs for a single HES ID from the study population.

We followed a target trial approach to mimic a randomized controlled trial that would be ideally conducted to estimate the causal programme impact as closely as possible (Supplementary Information section 1.3)^{55,56}. The population of interest consists of adults (aged 18 to 80 years) who received an HbA1c test between 1 January 2017 and 31 December 2018. Data were available until end of June 2020. Exclusion criteria were all patients who (1) exceeded the HbA1c threshold for diabetes or prediabetes before their index date (that is, date of their baseline HbA1c record) or (2) received any diabetes medication before their index date (all codes available in our Open Science Framework (OSF) repository). In a set of further analyses, we also excluded patients who had any HbA1c test before their index date to avoid any repeated HbA1c testing before treatment assignment. All results for this secondary cohort can be found in Supplementary Information section 3. Although specified in the NICE guideline as an alternative entry requirement, we did not use fasting plasma glucose as treatment assignment variable because routine testing of fasting plasma glucose to determine non-diabetic hyperglycaemia is rare compared to HbA1c testing⁵⁷.

Outcome and treatment definition

The primary outcome was glycaemic control measured as change in HbA1c between baseline and the final HbA1c taken during follow-up. Secondary outcomes included change in BMI, body weight, systolic and diastolic blood pressure, serum cholesterol concentrations (HDL cholesterol, LDL cholesterol and ratio between total and HDL cholesterol) and serum triglycerides concentration. We conducted exploratory analyses investigating the effect of programme entry onto probability of newly prescribed diabetes medications, blood pressure-lowering and/or lipid-lowering medication (evaluated separately), probability of any diabetes complication (ophthalmic, neurological or renal), all-cause mortality and emergency hospitalization for a MACE⁵⁸ during follow-up. The follow-up started at 6 months after the date of the baseline HbA1c record and continued until the date of an outcome or censoring event (such as death or transfer-out of the patient). Details on how each outcome was defined are shown in Supplementary Information section 1.1.

Receipt of treatment was captured as the record of a referral to a behaviour change programme or intensive lifestyle counselling during the 12 months after the baseline HbA1c test. Treatment primarily included referrals to the NHS DPP but we also included referrals to other structured programmes and intensive lifestyle counselling as they are likely to serve as an alternative when placement in NHS DPP is

not possible. A total of 26,970 patients with a baseline HbA1c between 42 and 47 mmol mol⁻¹ were referred to a behaviour change programme or intensive lifestyle counselling, of which 20,963 (77.7%) were referred to the NHS DPP. A total of 4,800 patients declined NHS DPP referrals offered by their GP. All records considered as treatments are listed in Supplementary Table 18. The NHS DPP guidelines stipulate that patients without a test indicating non-diabetic hyperglycaemia in their general practice records are not accepted into the programme (unless they have a history of gestational diabetes). However, patients may have been referred to an intensive lifestyle intervention that we considered to be comparable to the NHS DPP (for example, a structured weight loss programme) at the discretion of their GP (for example, due to other risk factors such as excess weight) or based on a second HbA1c test that exceeded the threshold within the referral window of 12 months. Thus, we performed a robustness analysis in which we restrict the referral window to 3 months after the baseline HbA1c measurement and only consider referrals to the NHS DPP as the treatment variable. This led to substantially fewer patients below the eligibility threshold being referred ($n = 620$). We observed similar treatment effects, with a comparable but slightly larger reduction in HbA1c compared to the primary analysis (Supplementary Table 19).

Main analysis

We used a regression discontinuity approach to estimate the effect of referral to intensive lifestyle counselling on our outcomes. The analysis consisted of two steps. First, we estimated the association between individual's baseline HbA1c and being referred to intensive lifestyle counselling ('first stage'). Second, we estimated the association between baseline HbA1c and each outcome by fitting separate regression lines above and below the HbA1c eligibility threshold. The difference in where these lines intersect the threshold quantifies the discontinuity in the outcome, our effect of interest.

Specifically, our analysis represents a fuzzy regression discontinuity (FRD) design, for which the treatment is not assigned deterministically but probabilistically. In the FRD design, we can estimate the effect of the patient presenting just above the eligibility threshold, that is, the effect of treatment eligibility as determined by the threshold rule or guideline itself. To obtain the effect of referral to intensive lifestyle counselling itself on those induced to accept a referral because of the threshold rule (so-called compliers), it is necessary to scale the effect of the threshold rule by the difference in the probability of treatment at the threshold. This results in a so-called CACE (or local average treatment effect), which is analogous to the intention-to-treat effect in a target trial (Supplementary Information section 1.3).

We used a local linear approach, which minimizes bias by limiting the study sample to a defined bandwidth around the threshold in which a linear regression can be estimated⁵⁹. The size of the bandwidth was automatically selected using a data-driven method that seeks to optimally balance the bias-variance trade-off⁵⁹. In addition, a triangular kernel was applied, such that individuals closer to the threshold were more heavily weighted than those further away. For computing confidence intervals, we used heteroskedasticity-robust standard errors. All tests for statistical significance in our regression discontinuity analyses are two-sided t -tests. Further analyses were performed to assess the robustness of results to bandwidth size, using a quadratic form and using a robust bias-corrected variance estimator for inferences⁶⁰.

We plotted the relationship between baseline HbA1c and referral to intensive lifestyle counselling to show the discontinuity in treatment assignment. We also present visual evidence in support of key identifying assumptions that can be tested in the data. The first is that the density of the data should be continuous around the threshold. This would be violated if patients (or GPs) could precisely manipulate the baseline HbA1c value. The second is that baseline covariates should be balanced (that is, continuous) at the threshold. As in randomized controlled clinical trials, evidence of balance on baseline observables

provides confidence that patients assigned to treatment and control conditions are exchangeable. We tested balance of baseline observables in a series of placebo tests in which we calculated local linear as well as global polynomial regressions over the entire support of the data to detect potential violations of this assumption.

Robustness checks

Our regression discontinuity approach achieves causal effect estimation through quasi-randomization rather than active adjustment for variables that may confound the association between treatment and outcome. Nevertheless, in robustness checks, we adjusted for variables that may be associated with outcomes to show robustness and improve precision of effect estimates⁶¹. These were age, gender, number of GP consultations during follow-up, time to follow-up record (that is, months between baseline and endline measurement) and prescription of relevant medications. For HbA1c, we adjusted results for diabetes medication prescription; for blood pressure, we adjusted for blood pressure-lowering medication prescription; and for cholesterol and triglycerides concentrations, we adjusted for lipid-lowering medication prescription, in particular statins, before the endline measure. Estimates for mortality and MACE hospitalization were adjusted for whether patients received any of these three types of medications before their death or first MACE hospitalization. We did further analyses to show that results are robust when restricting our sample to patients for whom we observe at least 12 and 18 months between baseline and endline measurement (Supplementary Tables 6 and 7).

All analyses were complete-case analyses and we assessed whether this could give rise to bias by evaluating if missingness in outcomes or time to follow-up changed discontinuously at the eligibility threshold (Supplementary Table 1). Following this, we performed a set of extra robustness checks following the approach by others to further investigate if there are any other inherent sources of bias or confounding³⁴. The core idea is "to repeat the experiment under conditions in which it is expected to produce a null result and verify that it does indeed produce a null result"³⁴. First, we tested a set of negative outcome control variables, that is, health outcomes for which we assumed that any causal effect through intensive lifestyle counselling (our proposed mechanism) would be impossible. Those variables are the probability of recording flu vaccination, pneumococcal vaccination, fracture, accidental injuries, routine cancer screenings (bowel, breast and cervical), consultation for skin abnormalities, asthma attacks, allergic reactions, hay fever, use of oral contraceptives, use of an intra-uterine device and onset of cancer or dementia. Second, we created a negative exposure control scenario, that is, we mimicked an analysis as closely as possible to our primary study design without the 'essential ingredient'—the NHS DPP programme. To do so, we leveraged the fact that NICE guidelines recommending monitoring and intensive lifestyle counselling for patients with an HbA1c of 42 mmol mol⁻¹ or higher had already been in place since late 2012, whereas the NHS DPP was only rolled out starting 2016. We performed a placebo analysis using an alternative patient cohort that is comparable in inclusion criteria to our primary analysis but who had baseline HbA1c measures in 2014 and 2015. During this period, individuals were already identified with prediabetes on the basis of the same threshold and public health guidance. However, the key difference was that there were limited intensive lifestyle programmes in place to which GPs could refer their patients. This led to a scenario in which potential confounders (for example, the effects of a diagnosis of prediabetes on patient and GP behaviours) could already be observed, whereas the proposed causal mechanism was 'deactivated'. We suggest that in this scenario, patients did not exhibit any significant health improvements.

All analyses were performed using RStudio and R statistical software (v.4.2.1)⁶². Mean-squared error (MSE) optimal bandwidths were automatically selected by the `rdbwselect` command of the `rdrobust` package⁶³. The two-sided significance level was set at 0.05. Blinding

and randomization of research participants is not actively performed in quasi-experimental study designs. We did not use statistical methods to predetermine sample size.

Difference-in-differences analysis

As further evidence, we present a difference-in-differences analysis leveraging the phased national roll-out of the NHS DPP programme in three waves (first wave start date, 1 June 2016; second wave start date, 1 April 2017; and third wave start date, 1 April 2018). Our approach is similar to the study design used by others who estimate the impact of the NHS DPP on type 2 diabetes incidence rates based on National Diabetes Audit data⁵⁰. Thanks to the differential timing of the NHS DPP for each wave, it is possible to compare patients from practices belonging to waves 1 and 2 with patients from practices belonging to wave 3 as the control group. To determine to which wave each CPRD practice belonged, we computed the share of eligible patients (patients scoring between 42 and 47 mmol mol⁻¹) who were referred in each roll-out phase for each practice. We considered practices to be participating in the NHS DPP in a given wave if they referred at least 3% of their eligible patients to the NHS DPP during that wave. Data were drawn from 1,383 practices (20.1% of all practices in England) covering more than 11 million registered patients. Of these practices, 188 were first wave, 433 second wave and 475 third wave practices based on the timing of their first recorded NHS DPP referral. We excluded 13 practices that already referred patients during a pilot phase before June 2016 and 274 practices that did not refer any patients at any time or closed before March 2020.

In our analysis, we treated April 2015 to May 2016 for the comparison of wave 1 to wave 3 practices and April 2015 to April 2017 for the comparison of wave 2 to wave 3 practices, as the pre-intervention period, capturing the trends before the implementation of the NHS DPP. We considered an implementation period to be 12 months because there is a lag between a practice enrolling in the NHS DPP, patients participating in the programme and notable effects of the 9 months' long programme on patients' health. Thus, the impact of the NHS DPP will be tested during the post-DPP introduction period. This post-period is April 2017 to March 2020 for the comparison of wave 1 to wave 3 practices and April 2018 to March 2020 for the comparison of wave 2 to wave 3 practices. We present group-time average treatment effects (with a 'group' being patients from GP practices in a given wave) for each relevant comparison as well as aggregated treatment effects that average effects in waves and across different lengths of exposure to the treatment (that is, implementation of the NHS DPP).

Specifically, we applied the framework by ref. 64 and used their accompanying R package *did*⁶⁵ for the case with repeated cross-sections data to account for variation in treatment timing and aggregate the several observed timepoints. We first estimated the group-time average treatment effect among the treated, $ATT(g, t)$, in each post-period, t , for wave 1 and wave 2, g , using wave 3 as control. The group-time ATT were estimated by regressing the individual-level HbA1c measure (latest available record in each period) on a treatment indicator (that is, whether the NHS DPP has already been implemented at the practice level), a time indicator for the current period and an interaction between those two indicators using ordinary least squares regression. The treatment effect was measured by the interaction term between the treatment indicator and the time indicator.

Next, we used different aggregation methods to summarize these group-time average treatment effects: a simple effects aggregation that reports the weighted average (by group size) of all available group-time average treatment effects, a group-specific effects aggregation that summarizes average treatment effects by the timing of the NHS DPP implementation and an event study aggregation. Event study aggregation uses a weighted average of the estimates grouped by length of time since NHS DPP implementation. We obtained standard errors clustered at the practice level and corresponding confidence

intervals for our estimates using the multiplier bootstrap procedures as described in ref. 64.

In line with changes in NICE guidelines¹⁴, the share of patients who have a recorded HbA1c test in each practice increased from 2015 to 2020. The increasing trends were similar across waves, except for wave 2 practices in 2019/2020 which were lower compared to wave 3 practices (Supplementary Fig. 11). Hence, as a robustness analysis, we applied a conditional parallel trends assumption; that is, we assume only that practices with the same share of patients having taken a HbA1c test would follow the same trend in HbA1c in the absence of treatment. This resulted in similar effect estimates (Extended Data Table 6b).

Instrumental variable estimation

To further triangulate evidence on the effect of the NHS DPP on glycaemic control, we present an analysis in which we use the regional variation in NHS DPP coverage as an instrumental variable for the actual receipt of programme referral.

Instrumental variable (IV) estimation is a statistical technique used to address endogeneity in regression analysis, for which the independent variable (in our case, being referred to the NHS DPP) is correlated with the error term, for example, due to selection bias in treatment assignment. The IV estimation requires several assumptions to be valid. First, the IV used in the analysis, that is, the regional variation in NHS DPP coverage, must be correlated with the endogenous variable of interest, that is, the patient-level probability of being referred to the NHS DPP. Second, the IV must be independent of the error term in the regression equation, meaning that it cannot be correlated with any other factors that affect the outcome variable. This assumption requires that the instrumental variable is not affected by any unmeasured confounding variables that are also affecting the outcome variable. Third, the IV must affect the outcome variable only through their effect on the endogenous variable and not through any other pathways. This assumption requires that the IV does not have any direct effect on the outcome variable. Last, the IV must have a monotonic effect on the endogenous variable.

We computed the regional NHS DPP coverage by dividing the number of patients registered with practices eligible for the DPP (which varies by programme wave) by the total number of patients registered in each region (Extended Data Fig. 2). The regional variable in CPRD refers to nine Strategic Health Authorities (SHAs), namely East Midlands, East of England, London, North East, North West, South Central, South East Coast, South West, West Midlands and Yorkshire and the Humber, which were part of the structure of the NHS in England between 2002 and 2013. We were able to match information that we received from the National Diabetes Audit on NHS DPP practice eligibility in each wave (practices listed by clinical commissioning groups, CCGs) to the SHAs through publicly available information on primary care trusts (PCTs)⁶⁶. In cases in which we found no match through PCTs, we matched the CCGs through output area, available from the Office for National Statistics^{67,68}. We constructed a panel dataset, consisting of yearly observations for all patients eligible for the NHS DPP from April 2016 to March 2020. Specifically, patients entered the panel in the year in which they first had an HbA1c test scoring between 42 and 47 mmol mol⁻¹ and were retained for each year during which HbA1c test results were available in their GP records. We applied the same inclusion criteria (aged 18 to 80 years, no previous use of diabetes medication and no history of diabetes) as for the regression discontinuity analysis.

We computed a cumulative treatment status for each year t ; that is, a patient was considered treated if receiving a referral by the end of year t and remained treated thereafter. We verified the first assumption of IV estimation by regressing the cumulative treatment status on the NHS DPP coverage in each region in the previous year $t - 1$ (first stage). We found that the regional NHS DPP coverage was significantly associated with the probability of being referred by the end of the following year (0.038, 95% CI 0.014, 0.063, $P < 0.001$).

We then used a 2SLS regression to estimate the effect of lagged treatment status on current HbA1c, with the lagged regional NHS DPP coverage serving as the instrumental variable for the lagged treatment status. We accounted for cases in which several HbA1c measurements were recorded for a patient within a year by averaging the measurements. We controlled for practice- and year-level fixed effects and clustered standard errors at the practice level. The resulting effect estimate represents the local average treatment effect, the causal effect of the endogenous treatment variable among those who would comply with the treatment if assigned to receive it. In other words, it measures the average effect of being referred to the NHS DPP for patients whose referral status changes only because of the variation in regional NHS DPP coverage, rather than because of their own characteristics or preferences. In addition, we report the intention-to-treat effect, which was estimated by regressing the lagged NHS DPP coverage directly onto current HbA1c. Overall, this approach allowed us to address potential endogeneity issues from conventional OLS regression and provide more reliable estimates of the treatment effect on HbA1c concentrations. We used the R package *lfe*⁶⁹ for this analysis.

Multivariate regression analyses, propensity score matching and panel regression

We also performed a set of extra regression-based analyses for our primary outcome HbA1c. On the basis of our target trial approach, these analyses included all patients eligible for the NHS DPP in our study period, that is, adults (aged 18 to 80 years) who had a HbA1c test greater than or equal to 42 mmol mol⁻¹ and less than 47 mmol mol⁻¹ between January 2017 and December 2018 with no previous use of diabetes medication and no history of diabetes.

First, we report unadjusted and adjusted associations between referral to the NHS DPP and endline HbA1c from linear regressions with robust standard errors clustered at the practice level and practice-fixed effects. In multivariate regression, we successively added (1) baseline variables that were available for the full sample, that is, baseline HbA1c measure, gender, age at baseline, average yearly number of GP consultations in 3 years before baseline, IMD, receipt of flu vaccination in previous 5 years, receipt of pneumococcal vaccination in previous 5 years, previous receipt of lifestyle advice, previous receipt of weight loss counselling, previous consultation for skin abnormalities and previous participation in bowel cancer screening, (2) baseline BMI and (3) post-baseline diabetes medication and time between baseline and endline HbA1c measurement (in months), as covariates.

Second, we used propensity score matching to estimate the average marginal effect of the NHS DPP referral on HbA1c accounting for confounding by the included covariates. We estimated propensity scores by performing a logistic regression of programme referral on the covariates and using a 1-to-1 nearest-neighbour propensity score matching without replacement (using the R package *MatchIt*⁷⁰). The covariates that we successively added to the matching procedure were (1) baseline HbA1c, age and gender, (2) number of average yearly GP consultations (in previous 3 years), (3) IMD and region, (4) receipt of previous lifestyle advice, (5) time between baseline and endline HbA1c measurement (in months) and (6) post-baseline diabetes medication. After matching, all standardized mean differences for the covariates were below 0.1 and all standardized mean differences for squares and two-way interactions between covariates were below 0.15, indicating adequate balance. We used *g*-estimation to estimate average treatment effects^{71,72}. Briefly, the first step is to specify a model for the outcome based on the treatment and covariates and potential outcomes are then estimated for each unit by computing predicted values for both treatment and control groups. The mean of estimated potential outcomes is computed for each group and the difference or ratio of these averages is used to estimate the treatment effect. Combining propensity score matching and *g*-computation has been shown to effectively minimize confounding bias⁷³. We used the R package *marginalEffects*⁷⁴

to implement *g*-computation with a linear model, treatment-covariate interactions, covariates and cluster-robust standard errors and estimated the average marginal effect of NHS DPP referral on HbA1c.

Third, using the same panel dataset as for the IV estimation described above, we estimated the one-year lagged treatment effect of programme referral on HbA1c from a regression with individual-fixed effects to account for time-invariant unobserved patient characteristics and year-fixed effects to control for the secular variation. As in the IV estimation, we accounted for cases in which several HbA1c measurements were recorded for a patient within a year by averaging the measurements. We used the R package *lfe*⁶⁹ to specify the linear model with several fixed effects and standard errors clustered at the practice level.

Heterogenous treatment effects

We present stratified regression discontinuity analyses for primary and secondary outcomes, using local linear regressions with automatically selected optimal bandwidths for each subgroup, in Supplementary Tables 8–16. Prespecified stratifying variables used in this analysis were gender (male versus female), age group (18–39, 40–59, 60–80 years), ethnicity (Asian, Black, mixed or other, white; based on the HES APC dataset), socioeconomic status and practice residency (rural versus urban). Socioeconomic status was based on the 2015 English IMD composite score, resulting in five quintiles ranging from 1 (least deprived) to 5 (most deprived) and mapped through postcode of residence for patients in English practices that have consented to participate in the linkage scheme. We further stratified our analyses for the following exploratory comparisons: having previously received general lifestyle advice versus not; having previously received counselling on weight loss versus not; having previously received a consultation for skin abnormalities versus not (that is, record, photography, excision, biopsy or removal skin lesion or mole); having previously participated in a bowel cancer screening activity versus not (that is, record of faecal occult blood testing or participation in NHS bowel cancer screening programme); having previously received a flu vaccination versus not; and having previously adhered to a statin prescription versus not. For the last, we defined a patient as non-compliant if the interval of prescriptions since their first prescription within 5 years before the baseline test exceeded 3 months. Statin prescriptions must generally be renewed every 2 months. All medical codes are available in our OSF repository.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

This study used data from the CPRD Aurum and NHS England HES APC database. The data are available from CPRD (<https://cprd.com>) but restrictions apply to the availability of these data, which were used under license for the current study and so are not publicly available. Owing to CPRD license restrictions, we are unable to share data.

Code availability

All medical codes and algorithms to define variables and R analysis code are available in the Supplementary Information or at the OSF repository (https://osf.io/rqz6x/?view_only=abc4c7a3abcb457596cec9fe2664f542).

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Acknowledgements This study is based on data from the Clinical Practice Research Datalink obtained under license from the UK Medicines and Healthcare products Regulatory Agency. The data are provided by patients and collected by the NHS as part of their care and support. The interpretation and conclusions contained in this study are those of the authors alone. This work was supported by the Alexander von Humboldt Foundation through the Alexander von Humboldt Professorship awarded to T.B. Data storage and computing resources used in this work were supported by the Ministry of Science, Research and the Arts Baden-Wuerttemberg, Germany, German Research Foundation, the state of Baden-Wuerttemberg, Germany and the German Research Foundation grant no. INST 35/1314-1 FUGG. P.G. was supported by the National Institute of Allergy and Infectious Diseases (1DP2AI171011) and the Chan Zuckerberg Biohub investigator award. J.M.L. acknowledges support from the German Academic Scholarship Foundation.

Author contributions J.M.L., C.B., A.J., J.I.D., S.V. and P.G. conceived of the research. T.B., S.V. and P.G. acquired funding for the research project. J.M.L., M.X. and F.M. curated the data. J.M.L., M.X., C.B. and P.G. designed the statistical analyses with consults from F.M., T.B. and S.V. J.M.L. and M.X. analysed the data. C.B. and P.G. supervised the analysis. J.M.L. and P.G. wrote the paper with edits from C.B., M.X., F.M., A.J., J.I.D., T.B. and S.V. All authors approved the final manuscript.

Competing interests The authors declare no competing interests.

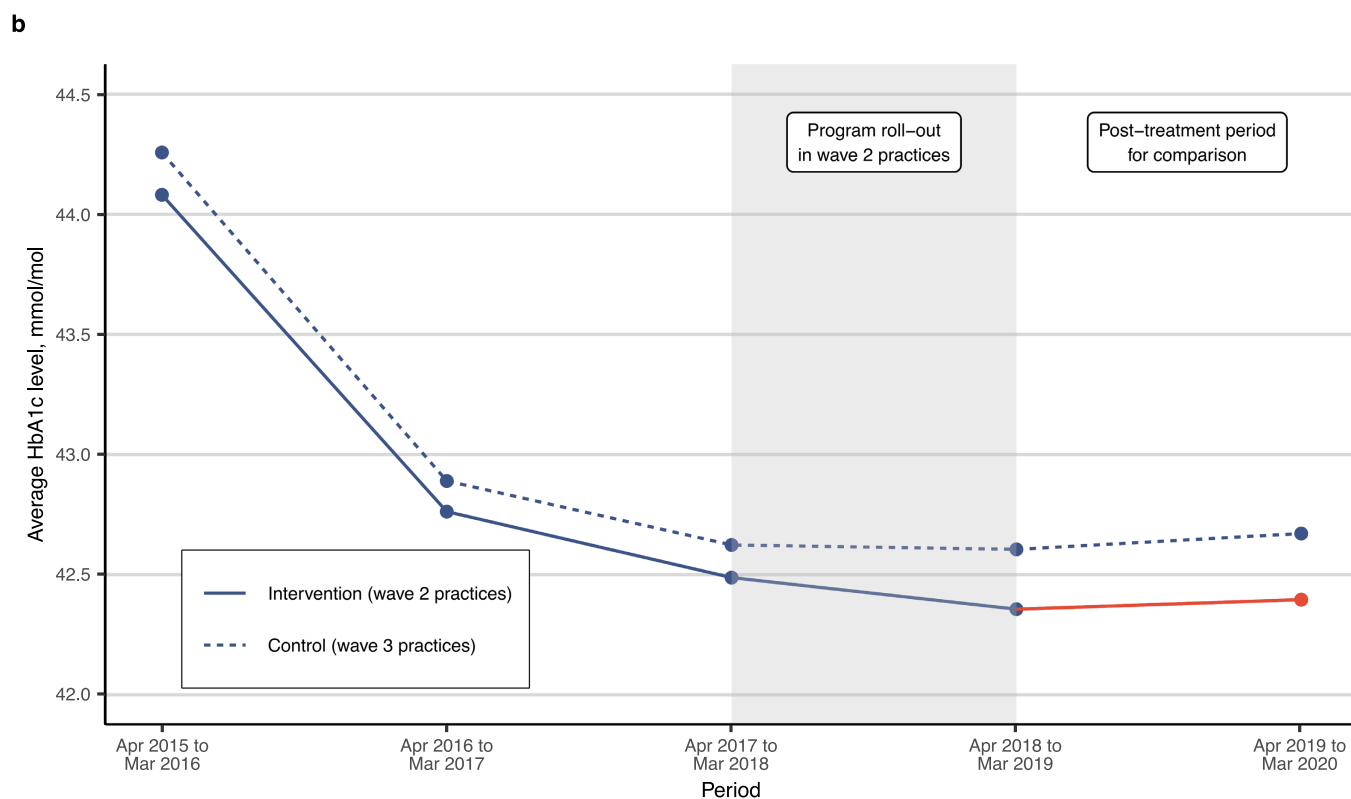
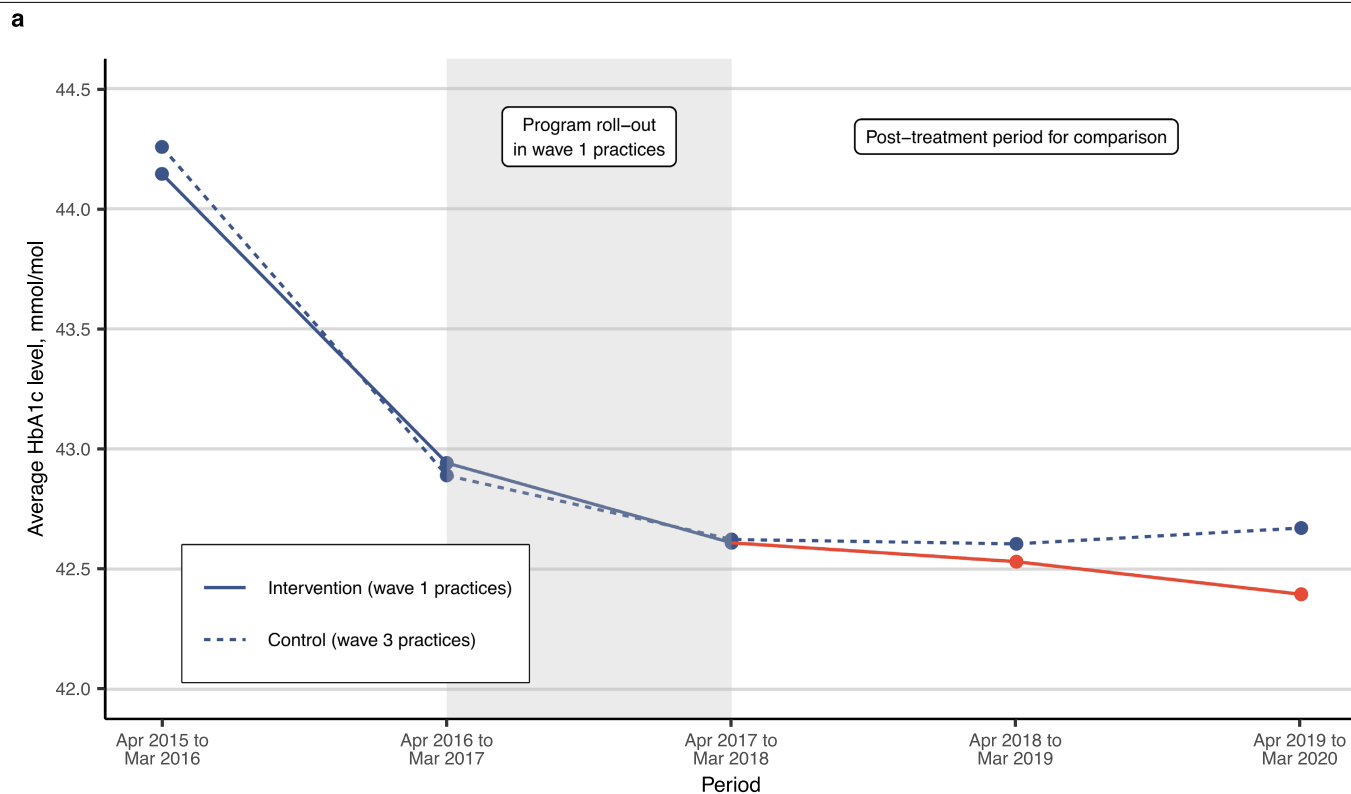
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-023-06756-4>.

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Peer review information *Nature* thanks Ziyad Al-Aly, Theis Lange and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

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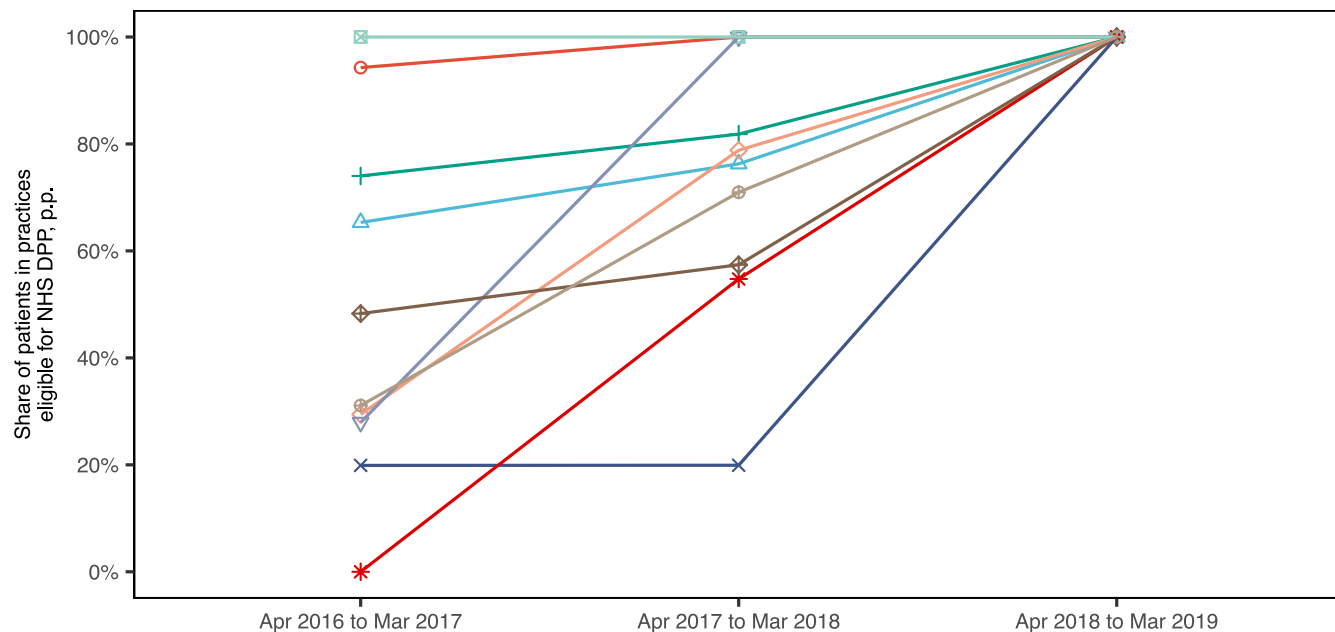


Extended Data Fig. 1 | Trends in glycated haemoglobin (HbA1c) before, during and after programme roll-out. Weighted average HbA1c in one-year intervals from April 2015 to March 2020, for (a) wave 1 and (b) wave 2 practices (intervention) compared to wave 3 practices (control). The y-axis does not start

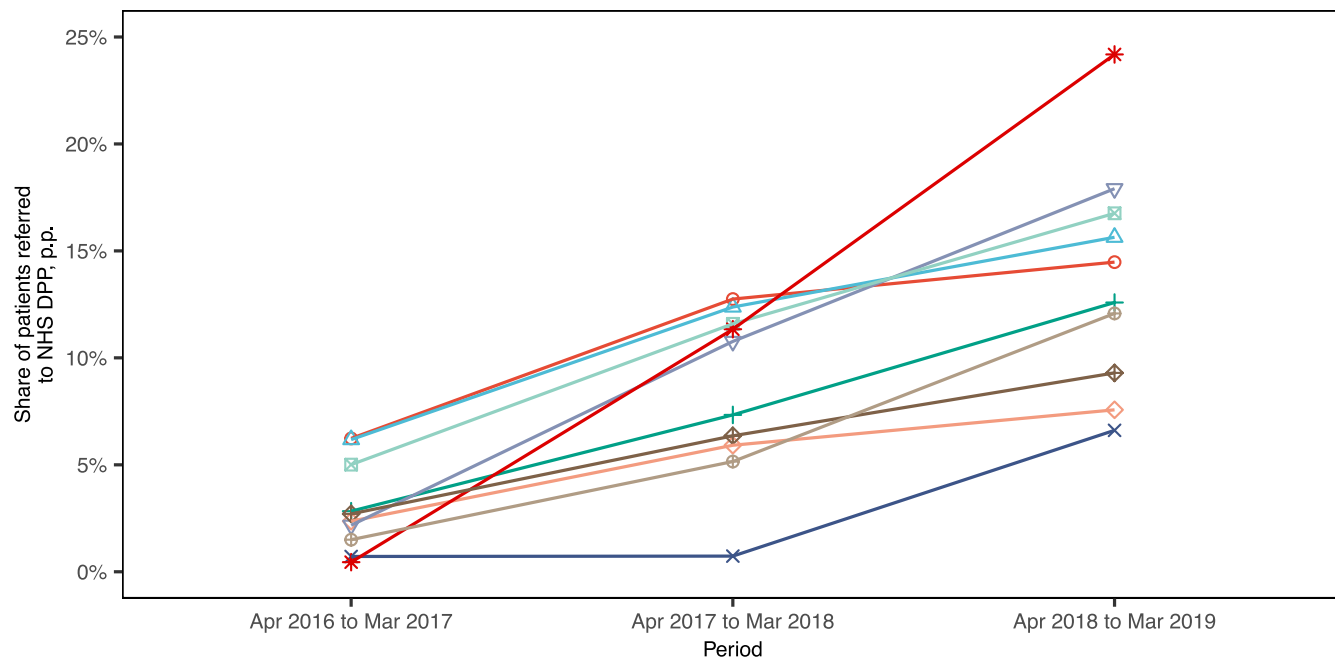
from 0, weighting by number of individuals for each practice, by year. The roll-out of the NHS DPP started in June 2016 for wave 1, in April 2017 for wave 2 and in April 2018 for wave 3.

Article

a Share of eligible patients



b Share of referred patients



—○— East Midlands
 —+— London
 —◇— North West
 —⊠— South East Coast
 —⬢— West Midlands
—△— East of England
—×— North East
—▽— South Central
—*— South West
—⊕— Yorkshire and the Humber

Extended Data Fig. 2 | Regional and temporal variation in NHS DPP programme implementation. (a) Share of patients eligible for NHS DPP derived via official practice eligibility by roll-out wave and (b) share of patients referred to NHS DPP in each of the nine Strategic Health Authorities.

Extended Data Table 1 | Sample characteristics

Variable	%	N
Age in years, mean (SD)	51 (16)	
18 to 29	11.6	243 793
30 to 39	14	295 282
40 to 49	19.5	411 201
50 to 59	21.9	461 006
60 to 69	17.9	377 073
70 to 80	15.1	318 021
Gender		
Female	56.5	915 717
Male	43.5	1 190 659
BMI, mean (SD)	28.1 (6.1)	
< 25	33.1	436 041
25 to 29.9	35.2	464 217
30 to 34.9	19.6	258 266
35 to 39.9	7.7	101 321
≥ 40	4.4	58 444
Hypertension stage based on systolic and diastolic blood pressure		
Normal (< 120 and 80)	19.3	188 906
Prehypertension (120-139 or 80-89)	59.6	583 428
Stage 1 Hypertension (140-159 or 90-99)	18.6	182 074
Stage 2 Hypertension (≥160 or ≥ 100)	2.5	24 222
	Mean	SD
Baseline blood pressure		
Diastolic, mmHg	80	11
Systolic, mmHg	132	18
Baseline lipids		
HDL, mmol/l	1.47	0.45
LDL, mmol/l	3.02	0.96
Total-cholesterol-to-HDL ratio	3.77	1.25
Triglycerides, mmol/l	1.55	0.97
Number of yearly GP consultations	12.4	10.8
Number of prior emergency hospitalization	1.13	5.37

Description of the primary cohort including information on age, gender, body mass index (BMI), hypertension stage, blood pressure, serum lipids, number of yearly general practitioner (GP) consultations and number of previous emergency hospitalization at baseline. Age, gender and number of yearly GP consultations was available for the full sample. The primary cohort consists of 2,106,376 patients. 2 052 480 of these (97.4%) had been registered with their GP for at least 6 months following the index date. The primary outcome of endline HbA1c was available for 1 043 268 patients (50.8% of 2 052 480, table S1) and the median time to endline HbA1c during follow-up was 20.5 months (interquartile range, 13.5-26.8). Of the 2 052 480 patients who had a follow-up time of at least 6 months, 2 037 384 (99.3%) were linkable to HES hospitalization data. Missingness in baseline and endline BMI, blood pressure and lipids are shown in table S1. Exchangeability of patients below and above the threshold are demonstrated in placebo tests with baseline characteristics (balance tests) in Extended Data Table 2. The secondary cohort is described in the Supplementary Information S4.1.

Article

Extended Data Table 2 | Placebo tests with baseline characteristics (Balance tests)

Variable at baseline*	Primary cohort		Secondary cohort	
	Estimate (95% CI)	P value	Estimate (95% CI)	P value
Age in years [‡]	0.22 (-0.13, 0.57)	0.225	0.255 (-0.085, 0.595)	0.141
Female, p.p.	-0.1 (-0.7, 0.6)	0.860	0.1 (-0.7, 0.8)	0.866
Baseline BMI	-0.03 (-0.12, 0.07)	0.603	-0.03 (-0.14, 0.08)	0.566
Baseline weight, kg	0.02 (-0.27, 0.31)	0.880	-0.05 (-0.39, 0.30)	0.794
Average yearly number of GP consultations [‡]	-0.05 (-0.18, 0.08)	0.462	0.07 (-0.08, 0.21)	0.371
Baseline blood pressure, mmHg				
Diastolic	0.053 (-0.25, 0.35)	0.729	-0.13 (-0.49, 0.24)	0.486
Systolic	0.16 (-0.02, 0.34)	0.083	0.05 (-0.17, 0.27)	0.655
Baseline lipids				
HDL, mmol/l	0 (-0.006, 0.006)	0.969	0.009 (0.002, 0.016)	0.008
LDL, mmol/l	-0.004 (-0.022, 0.014)	0.672	-0.023 (-0.045, 0)	0.045
Total-cholesterol-to-HDL ratio	-0.002 (-0.022, 0.017)	0.826	-0.031 (-0.055, -0.007)	0.010
Triglycerides, mmol/l	0.006 (-0.009, 0.022)	0.409	0 (-0.018, 0.019)	0.961
Number of flu vaccinations [‡]	-0.01 (-0.03, 0)	0.054	-0.01 (-0.02, 0.01)	0.314
Flu vaccination [‡] , p.p.	-0.3 (-0.7, 0.1)	0.107	-0.1 (-0.5, 0.4)	0.787
Pneumococcal vaccination [‡] , p.p.	0.1 (-0.2, 0.3)	0.604	0 (-0.3, 0.2)	0.899
Number of lifestyle advice [‡]	-0.01 (-0.02, 0.01)	0.269	-0.01 (-0.02, 0.01)	0.424
Lifestyle advice [‡] , p.p.	-0.1 (-0.6, 0.5)	0.796	0.1 (-0.5, 0.7)	0.771
Weight loss counseling [‡] , p.p.	0.1 (-0.2, 0.3)	0.436	0 (-0.3, 0.2)	0.887
Routine screening [‡] , p.p.				
Colorectal [‡]	0.6 (-0.7, 1.9)	0.376	0.3 (-0.9, 1.5)	0.642
Skin	0.2 (-0.2, 0.7)	0.316	0.2 (-0.3, 0.7)	0.517
Breast [‡]	1.4 (-0.5, 3.2)	0.142	1.7 (0, 3.4)	0.050
Statin adherence [‡] , p.p.	0.1 (-0.6, 0.9)	0.720	0.2 (-0.8, 1.2)	0.713

*Missingness in baseline variables is shown in Supplementary Table S2.

[‡]The pre-period refers to five years prior to the baseline HbA1c test.

[‡]For age, colorectal screening and breast cancer screening, we assumed a quadratic relationship with baseline HbA1c.

BMI = body mass index. GP = General practitioner. P.p. = percentage points. This table shows placebo tests that demonstrate exchangeability of patients above and below the eligibility threshold. We used local linear regression with heteroskedasticity-robust standard errors and triangular kernel weights, within the bandwidth used in our primary analysis. For gender, vaccinations, lifestyle advice, weight loss counselling and statin adherence, effect estimates can be interpreted as the difference in percentage points at the threshold. The results for routine breast screening are restricted to women only. The results for statin adherence are restricted to patients with a record of statin prescription in the pre-period.

Extended Data Table 3 | Regression discontinuity results of being eligible for and of being referred to, intensive lifestyle counselling

Outcome	Unadjusted				Adjusted for medication*		MSE-optimal BW	Sample size†
	Effect of being eligible		Effect of referral		Effect of referral			
	Estimate (95% CI)	P value	Estimate (95% CI)	P value	Estimate (95% CI)	P value		
Primary outcome								
HbA1c, mmol/mol	-0.10 (-0.16, -0.03)	0.006	-0.85 (-1.46, -0.24)	0.006	-0.94 (-1.55, -0.34)	0.002	3.8	298 822
Secondary outcomes								
BMI, kg/m²	-0.15 (-0.21, -0.09)	< 0.001	-1.35 (-1.88, -0.83)	< 0.001	-	-	3.8	184 087
Weight, kg	-0.33 (-0.48, -0.18)	< 0.001	-2.99 (-4.38, -1.61)	< 0.001	-	-	3.4	208 111
Diastolic BP, mmHg‡	-0.16 (-0.4, 0.07)	0.178	-1.35 (-3.31, 0.61)	0.178	-1.54 (-3.44, 0.35)	0.111	7.5	211 778
Systolic BP, mmHg‡	-0.24 (-0.6, 0.11)	0.175	-2.03 (-4.96, 0.91)	0.176	-2.41 (-5.21, 0.39)	0.092	5.6	243 292
HDL cholesterol, mmol/l§	0 (0, 0.01)	0.033	0.04 (0, 0.09)	0.033	0.04 (0, 0.09)	0.034	6.3	290 932
LDL cholesterol, mmol/l§	0.01 (-0.01, 0.04)	0.265	0.11 (-0.08, 0.3)	0.265	0.04 (-0.12, 0.21)	0.596	5.7	107 790
Total-cholesterol-to-HDL ratio§	0 (-0.02, 0.02)	0.741	-0.03 (-0.20, 0.14)	0.741	-0.09 (-0.25, 0.08)	0.286	5.4	178 852
Triglycerides, mmol/l§	-0.04 (-0.06, -0.01)	0.002	-0.33 (-0.54, -0.12)	0.002	-0.32 (-0.52, -0.12)	0.002	4.4	126 297
Exploratory outcomes								
Diabetes diagnosis, RD	0.39 (0.21, 0.57)	< 0.001	3.66 (2.00, 5.33)	< 0.001	-	-	3.1	327 756
Diabetes medication, RD	0.30 (0.18, 0.41)	< 0.001	2.72 (1.66, 3.78)	< 0.001	-	-	3.5	501 241
Antihypertensive medication, RD‡	0.11 (-0.38, 0.6)	0.660	0.93 (-3.22, 5.09)	0.659	-	-	4.5	390 878
Lipid-lowering medication, RD§	0.29 (-0.23, 0.82)	0.271	2.61 (-2.03, 7.25)	0.271	-	-	3.2	344 522
Diabetes complication, RD	0.06 (-0.07, 0.19)	0.344	0.58 (-0.62, 1.78)	0.344	0.15 (-0.98, 1.28)	0.797	3.3	493 431
Mortality, RD	0.05 (-0.09, 0.20)	0.471	0.47 (-0.81, 1.76)	0.471	0.67 (-0.62, 1.95)	0.307	4.9	698 224
Emergency MACE hospitalization, RD	-0.05 (-0.23, 0.12)	0.550	-0.48 (-2.05, 1.09)	0.550	-0.12 (-1.68, 1.45)	0.883	7.5	211 778

*Effects for HbA1c and diabetes complication were adjusted for diabetes medication prescription; effects for lipid levels were adjusted for lipid-lowering medication prescription; effects for diastolic and systolic blood pressure were adjusted for blood pressure-lowering medication; and effects for mortality and MACE hospitalization were adjusted for all three medication groups. All relevant medications are listed in our Open Science Framework project (see code availability statement).

†Sample size within MSE-optimal bandwidth.

‡Sample restricted to those without prior lipid-lowering medication prescription. At baseline, 411 288 (20.0%) people had already received at least one prescription for a lipid-lowering medication.

§Sample restricted to those without prior blood pressure-lowering medication prescription. At baseline, 749 884 (36.5%) people had already received at least one prescription for a blood pressure-lowering medication.

BMI = Body mass index. BP = Blood pressure. MACE = Major adverse cardiovascular event. RD = Risk difference (i.e., difference in the probability of the outcome in percentage points). This table shows the main regression discontinuity results for the primary cohort including effect size estimate, 95% confidence intervals (95% CI), P values, mean-squared error (MSE) optimal bandwidth (BW) and sample size. The effects were estimated in local linear regressions with heteroskedasticity-robust standard errors and triangular kernel weight and evaluated using two-sided t-tests ($p < 0.05$). The definition of all outcomes is detailed in the Supplementary Information 1.2. Additional details are available in Methods ('Main Analysis'). We compared statistical significance to results using robust bias-corrected confidence intervals, which yielded the same statistical inferences except for diabetes medication, which was no longer significant (Extended Data Table 4). Results for the secondary cohort are available in the Supplementary Information S4.

Extended Data Table 4 | Regression discontinuity results with robust bias-corrected confidence intervals of being eligible for and of being referred to, intensive lifestyle counselling

Outcome	Unadjusted				Adjusted for medication*		MSE-optimal BW	Sample size†
	Effect of being eligible		Effect of referral		Effect of referral			
	Estimate (Rob. 95% CI)	P value	Estimate (Rob. 95% CI)	P value	Estimate (Rob. 95% CI)	P value		
Primary outcome								
HbA1c, mmol/mol	-0.10 (-0.35, -0.07)	0.004	-0.85 (-3.26, -0.69)	0.003	-0.94 (-3.3, -0.76)	0.002	3.8	298 822
Secondary outcomes								
BMI, kg/m²	-0.15 (-0.28, -0.02)	0.029	-1.35 (-2.59, -0.23)	0.019	-	-	3.8	184 087
Weight, kg	-0.33 (-0.58, 0.08)	0.133	-2.99 (-5.59, 0.46)	0.097	-	-	3.4	208 111
Diastolic BP, mmHg‡	-0.16 (-0.52, 0.17)	0.328	-1.35 (-4.31, 1.38)	0.314	-1.54 (-4.47, 1.02)	0.219	7.5	211 778
Systolic BP, mmHg‡	-0.24 (-0.74, 0.28)	0.371	-2.03 (-6.17, 2.24)	0.360	-2.41 (-6.24, 1.66)	0.255	5.6	243 292
HDL cholesterol, mmol/l§	0 (0, 0.01)	0.031	0.04 (0.01, 0.14)	0.024	0.04 (0.01, 0.14)	0.025	6.3	290 932
LDL cholesterol, mmol/l§	0.01 (-0.01, 0.06)	0.209	0.11 (-0.1, 0.51)	0.195	0.04 (-0.15, 0.37)	0.414	5.7	107 790
Total-cholesterol-to-HDL ratio§	0 (-0.03, 0.03)	0.970	-0.03 (-0.29, 0.27)	0.961	-0.09 (-0.34, 0.21)	0.658	5.4	178 852
Triglycerides, mmol/l§	-0.04 (-0.11, -0.03)	0.001	-0.33 (-1.06, -0.32)	< 0.001	-0.32 (-1.05, -0.34)	< 0.001	4.4	126 297
Exploratory outcomes								
Diabetes diagnosis, RD	0.39 (-0.46, 0)	0.046	3.66 (-3.67, 0.6)	0.160	-	-	3.1	327 756
Diabetes medication, RD	0.30 (-0.06, 0.37)	0.152	2.72 (-0.35, 3.58)	0.107	-	-	3.5	501 241
Antihypertensive medication, RD‡	0.11 (-0.69, 1.01)	0.718	0.93 (-5.84, 8.61)	0.706	-	-	4.5	390 878
Lipid-lowering medication, RD§	0.29 (-0.63, 1.47)	0.434	2.61 (-5.45, 13.14)	0.417	-	-	3.2	344 522
Diabetes complication, RD	0.06 (-0.09, 0.19)	0.677	0.58 (-2.89, 1.94)	0.698	0.15 (-3.11, 1.66)	0.550	3.3	493 431
Mortality, RD	0.05 (-0.3, 0.19)	0.664	0.47 (-2.69, 1.76)	0.683	0.67 (-2.66, 1.8)	0.705	4.9	698 224
Emergency MACE hospitalization, RD	-0.05 (-0.42, 0.19)	0.460	-0.48 (-3.82, 1.68)	0.446	-0.12 (-3.76, 1.73)	0.468	7.5	211 778

*Effects for HbA1c and diabetes complication were adjusted for diabetes medication prescription; effects for lipid levels were adjusted for lipid-lowering medication prescription; effects for diastolic and systolic blood pressure were adjusted for blood pressure-lowering medication; and effects for mortality and MACE hospitalization were adjusted for all three medication groups. All relevant medications are listed in our Open Science Framework project (see code availability statement).

†Sample size within MSE-optimal bandwidth.

‡Sample restricted to those without prior lipid-lowering medication prescription. At baseline, 411 288 (20.0%) people had already received at least one prescription for a lipid-lowering medication.

§Sample restricted to those without prior blood pressure-lowering medication prescription. At baseline, 749 884 (36.5%) people had already received at least one prescription for a blood pressure-lowering medication.

BMI = Body mass index. BP = Blood pressure. MACE = Major adverse cardiovascular event. RD = Risk difference (i.e., difference in the probability of the outcome in percentage points). Rob. = Robust. This table shows the main regression discontinuity results for the primary cohort including effect size estimate, robust bias-corrected 95% confidence intervals (95% CI), P values, mean-squared error (MSE) optimal bandwidth (BW) and sample size. The effects were estimated in local linear regressions with triangular kernel weight and evaluated using two-sided t-tests (p < 0.05). The definition of all outcomes is detailed in the Supplementary Information 1.2. Additional details are available in Methods ('Main Analysis').

Extended Data Table 5 | Negative outcome controls

Outcome	Bandwidth	Estimate (95% CI)	P value	Sample size
Flu vaccination	3.1	0.10 (-0.43, 0.63)	0.714	513 473
Number of flu vaccinations*	3.1	0.22 (-0.85, 1.28)	0.689	480 703
Pneumococcal vaccination	4.2	-0.11 (-0.27, 0.05)	0.184	720 264
Fracture	5.2	-0.05 (-0.2, 0.11)	0.537	946 230
Accidental injuries	5.9	0.05 (-0.11, 0.21)	0.545	946 230
Number of accidental injuries*	6.1	0.18 (-0.12, 0.48)	0.241	117 3602
Routine screenings				
Colorectal	2.3	-0.41 (-1.2, 0.37)	0.303	513 473
Breast	2.8	-0.79 (-1.84, 0.26)	0.141	173 177
Cervical	2.8	0.15 (-0.81, 1.11)	0.759	173 177
Skin abnormality	4.6	-0.48 (-0.79, -0.16)	0.003	720 264
Allergic reaction	5.8	-0.01 (-0.17, 0.16)	0.950	946 230
Contraceptive use				
IUD	3.8	0.04 (-0.15, 0.24)	0.662	267 133
Oral	2.1	-0.24 (-0.89, 0.4)	0.459	173 177
Short-term onset of†				
Cancer	5.5	0.14 (-0.02, 0.31)	0.077	889 887
Dementia	6.5	-0.02 (-0.09, 0.05)	0.547	1 166 480

*Effect estimate can be interpreted as the difference in number of events during the follow-up period at the threshold.

†Sample restricted to those without any previous diagnosis of cancer or dementia, respectively. While healthy lifestyle behaviours are assumed to be protective factors for cancer and dementia onset, the median follow-up time (29 months) is sufficiently short to assume that any plausible effects are negligible.

IUD = Intra-uterine device. Regression coefficients from local linear regression with MSE-optimal bandwidth, 95% confidence intervals (95% CI), P values and sample size. This table shows the main regression discontinuity analysis (primary cohort) for health outcomes which we did not expect to be plausibly affected by referral to the NHS DPP ('negative outcome controls'). Effect estimates can be interpreted as the difference in percentage points at the threshold. The results for routine breast and cervical cancer screening as well as contraceptive use are restricted to women only.

Extended Data Table 6 | Aggregated treatment effect estimates of NHS DPP implementation on HbA1c from difference-in-differences analysis

Partially aggregated (95% CI), mmol/mol				Single parameters (95% CI), mmol/mol
<i>(a) Unadjusted</i>				
Simple weighted average				-0.144 (-0.188, -0.100)
Group-specific effects	<u>Wave 1</u>	<u>Wave 2</u>		
	-0.173 (-0.253, -0.094)	-0.126 (-0.196, -0.057)		
Event study	<u>Year 1</u>	<u>Year 2</u>	<u>Year 3</u>	
	-0.099 (-0.163, -0.036)	-0.136 (-0.205, -0.067)	-0.328 (-0.433, -0.223)	-0.188 (-0.235, -0.141)
<i>(b) Adjusted for share of patients with a recorded HbA1c test</i>				
Simple weighted average				-0.155 (-0.196, -0.113)
Group-specific effects	<u>Wave 1</u>	<u>Wave 2</u>		
	-0.103 (-0.185, -0.021)	-0.187 (-0.251, -0.122)		
Event study	<u>Year 1</u>	<u>Year 2</u>	<u>Year 3</u>	
	-0.119 (-0.183, -0.055)	-0.165 (-0.229, -0.100)	-0.245 (-0.354, -0.136)	-0.176 (-0.227, -0.126)

This table reports aggregated treatment effect parameters under **(a)** the unconditional parallel trend assumption and the **(b)** conditional parallel trend assumption, adjusted for the practice-level share of patients with a recorded HbA1c test in each year. The row ‘Simple weighted average’ reports the weighted average (by group size) of all available group–time average treatment effects. The row ‘Group-specific effects’ summarizes average treatment effects by the timing of the NHS DPP implementation. The row ‘Event study’ reports average treatment effects by the length of exposure to the NHS DPP implementation. The column ‘Single parameters’ represents a further aggregation of each type of parameter as discussed in Callaway and Sant’Anna (2021).

Extended Data Table 7 | Overview of estimated effects of intensive lifestyle counselling on glycemic control

Analysis	Name of estimand	Interpretation	Sample	Change in HbA1c, mmol/mol		Location of details
				Estimate (95% CI)	P value	
Regression discontinuity	Effect of being eligible	Causal effect for those who are eligible for intensive lifestyle counseling as determined by the clinical guideline (irrespective of actual referral or adherence)	Primary cohort*	-0.10 (-0.16, -0.03)	0.006	Methods 'Main Analysis'; Extended Data Tables 3-4; Supplementary Tables S2-3
			Secondary cohort†	-0.15 (-0.24, -0.05)	0.002	
	Effect of referral (Complier average causal effect)	Causal effect for those who are referred to intensive lifestyle counseling because they crossed the eligibility threshold (i.e., may be conceptualized as the "intent-to-treat" effect in a target trial, irrespective of adherence)	Primary cohort*	-0.85 (-1.46, -0.24)	0.006	Methods 'Main Analysis'; Extended Data Tables 3-4; Supplementary Tables S2-3
			Secondary cohort†	-1.26 (-2.06, -0.46)	0.002	
Difference-in-differences	Simple weighted average	Weighted average of all available group-time treatment effects, grouped by group size	Eligible patients‡ in roll-out practices	-0.14 (-0.19, -0.10)	< 0.001	Methods 'Difference-in-differences analysis'; Extended Data Table 6; Extended Data Figure 1
	Event-aggregated average	Weighted average of all available group-time treatment effects, grouped by length of time since DPP implementation		-0.19 (-0.24, -0.14)	< 0.001	
	Event-specific effect at 3-year post-DPP introduction	Effect of comparing wave 1 practices to control practices during the last available observation period from April 2019 to March 2020		-0.33 (-0.44, -0.22)	< 0.001	
Instrumental variable	Effect of being eligible (Intent-to-treat effect in a 2SLS regression)	Effect of being eligible due to a change in program coverage in their region (irrespective of actual referral or adherence)	Eligible patients‡ from 2016 to 2020	-0.14 (-0.29, -0.001)	0.048	Methods 'Instrumental variable estimation'; Supplementary Table S17; Extended Data Figure 1
	Effect of referral (Complier average causal effect in 2SLS regression)	Effect for those who changed from not being referred to being referred due to a change in program coverage in their region		-3.77 (-7.52, -0.01)	0.049	
Individual fixed-effect regression	Treatment coefficient	One-year lagged effect of DPP referral from a regression with individual-fixed effects to account for time-invariant unobserved patient characteristics, and year-fixed effects to control for the secular variation	Eligible patients‡ in 2017 and 2018	-0.62 (-0.74, -0.51)	< 0.001	Methods 'Multivariate regression analyses, propensity score matching, and panel regression'; Supplementary Table S17
Propensity score matching	Fully adjusted average marginal effect (treatment coefficient via <i>g</i> computation)	Average marginal effect of DPP referral, matched based on and adjusted for baseline and post-baseline variables		-0.32 (-0.47, -0.16)	< 0.001	
Multivariable regression	Fully adjusted treatment coefficient	Association between DPP referral and endline HbA1c, adjusted for baseline and post-baseline variables		-0.33 (-0.46, -0.20)	< 0.001	

*The primary cohort consists of adult patients (aged 18 to 80 years) who had a recorded HbA1c measure between January 1st, 2017 and December 31st, 2018. We excluded all patients who exceeded the HbA1c threshold for diabetes or prediabetes (42 mmol/mol) prior to their index date (i.e., date of their baseline HbA1c record), or received any diabetes medication prior to their index date. The endline HbA1c measure must have been recorded at least six months after the baseline test.

†As for the secondary cohort, we additionally excluded patients who had any HbA1c measure (whether it exceeded the prediabetes threshold or not) prior to their index date

‡Eligible patients are all adult patients who had a HbA1c measure between 42 and 47 mmol/mol in the specified period.

This table shows the estimated effect of intensive lifestyle counselling (i.e., eligibility for, referral to, or implementation of the NHS DPP) on glycemic control (change in HbA1c in mmol/mol) from all analyses presented in this article. Details for each methodology can be found in the Methods (see column 'Location of Details').

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- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	The manuscript did not utilize any software for data collection. Rather, the analysis is based on electronic health record data routinely collected in English primary care practices (CPRD Aurum contains data from practices using EMIS Web® electronic patient record system) and hospitals (Hospital Episodes Statistics are informed by NHS healthcare provider systems).
Data analysis	RStudio with R version 4.2.1 All packages including date for version control are included in the R code. All medical codes and algorithms to define variables, and R analysis code are available in the Supplementary Information or at the Open Science Framework repository (https://osf.io/rqz6x/?view_only=abc4c7a3abcb457596cec9fe2664f542).

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- Accession codes, unique identifiers, or web links for publicly available datasets
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The manuscript used data from the Clinical Practice Research Datalink (CPRD) Aurum and the NHS England Hospital Episode Statistics Admitted Patient Care (HES APC) database. The data that support the findings of this study are available from CPRD (<https://cprd.com>), but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Due to CPRD license restrictions, we are unable to share data.

Human research participants

Policy information about [studies involving human research participants and Sex and Gender in Research](#).

Reporting on sex and gender

The exploratory gender-based analyses in our study are based on the gender variable as recorded in the CPRD electronic health records (documented by practitioners of general practices). While we acknowledge that these records may not fully reflect the gender identity of the participants as the variable is presently limited to female/male, we consider gender to be the more appropriate term for our study.

Population characteristics

The samples are drawn from adults registered with general practices in England. We present their characteristics including gender, age, and baseline physiological characteristics in Extended Data Table 1. Stratifying variables used in this study were gender (Male vs. female), age group (18-39, 40-59, 60-80), ethnicity (Asian, black, mixed or other, white; based on HES APC dataset), socioeconomic status, and practice residency (rural vs. urban). Socioeconomic status is based on the 2015 English Index of Multiple Deprivation composite score, resulting in five quintiles ranging from 1 (= "least deprived") to 5 (= "most deprived") and mapped via postcode of residence for patients in English practices that have consented to participate in the linkage scheme.

Recruitment

Does not apply (routine electronic health records)

Ethics oversight

The CPRD has been granted generic ethics approval for observational studies that make use of only anonymised data and linked anonymised NHS health-care data (Multiple Research Ethics Committee ref. 05/MRE04/87). All CPRD studies require scientific approval from the MHRA Independent Scientific Advisory Committee (ISAC). Our study protocol (20_000052) was approved in accordance with the Declaration of Helsinki.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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All studies must disclose on these points even when the disclosure is negative.

Study description

Quantitative quasi-experimental

Research sample

The study used data from the Clinical Practice Research Datalink (CPRD) Aurum and the NHS England Hospital Episode Statistics Admitted Patient Care (HES APC). CPRD Aurum is a large primary care database of de-identified electronic health records from a network of General Practitioner (GP) practices across the UK. CPRD is representative of the broader English population in terms of geographical spread, deprivation, age, and gender.

The primary analytical cohort for our regression discontinuity analysis consisted of adult patients, aged 18 to 80 years, registered with a GP practice and who received an HbA1c test between January 1st, 2017, and December 31st, 2018. Data were available until end of June 2020. We only included patients who (i) did not exceed the HbA1c threshold for diabetes or prediabetes prior to their index date (i.e., date of their baseline HbA1c record), and (ii) did not receive any diabetes medication prior to their index date (all codes available in our OSF repository). We chose this study sample as our primary analytical cohort as these patients most closely represent the target population of diabetes prevention efforts (individuals being tested for HbA1c; scoring below and above the prediabetes threshold defining high risk of progression to Type 2 diabetes, and thus, eligibility for the NHS Diabetes Prevention Programme). It mimics the population that would be ideally included in a target trial (see Supplemental Materials S1.3).

In a set of additional analyses, we only included patients who had their first-ever HbA1c test during the inclusion period to avoid any repeated HbA1c testing before treatment assignment (our secondary analytical cohort). We chose this study sample as our secondary analytical cohort as patients newly tested for HbA1c are most likely to benefit from being referred to intensive lifestyle counseling and, thus, represent a relevant population for targeted prevention efforts. For our placebo regression discontinuity analysis, the same exclusion criteria applied to patients who received an HbA1c test between January 1st, 2014, and December 31st, 2015.

For the difference-in-differences analyses, the observation period was extended to April 1, 2015 - March 31, 2020. Specifically, we compared patients' HbA1c record from practices belonging to waves 1 and 2 with patients from practices belonging to wave 3 as the control group. Data were drawn from 1,383 CPRD practices. Of these practices, 188 were first wave, 433 second wave, and 475 third wave practices based on the timing of their first recorded NHS DPP referral. We excluded 13 practices that already referred patients during a pilot phase before June 2016 and 274 practices that did not refer any patients at any time or closed before March 2020.

For the instrumental variable estimation and panel regression, we constructed a panel dataset, consisting of yearly observations for all patients eligible for the NHS DPP from April 2016 to March 2020. Specifically, patients entered the panel in the year in which they first had an HbA1c test scoring between 42 and 47 mmol/mol and were retained for each year during which HbA1c test results were available in their GP records. We applied the same inclusion criteria (aged 18 to 80 years, no prior use of diabetes medication and no history of prediabetes or diabetes) as for the regression discontinuity analysis. For the (matched) multivariate regression, we used the primary analytical cohort for the regression discontinuity analysis.

Sampling strategy	Due to the quasi-experimental/retrospective analysis strategy of routine patient data, sampling procedure is not an applicable category. All patients who met the inclusion criteria were retrospectively included. We refrained from doing a formal sample size calculation because key parameters (e.g. compliance with the threshold rule, missingness in outcomes, distribution of HbA1c measurements around the threshold) were unknown a priori and the prospective sample size (> 2 million) was unlikely to generate relevant constraints for the precision of the analysis of the main outcomes in the study.
Data collection	The manuscript did not rely on primary data collection. Rather, the analysis is based on electronic health records routinely collected in English primary care practices (CPRD Aurum contains data from practices using EMIS Web® electronic patient record system software) and hospitals (Hospital Episodes Statistics are informed by NHS healthcare provider systems).
Timing	Regression discontinuity/regression analysis: January 1, 2017 - June 30, 2020 Difference-in-differences analysis: April 1, 2015-March 31, 2020 Instrumental variable estimation/panel regression analysis: April 1, 2020 to March 31, 2020 Placebo analysis: January 1, 2014-June 30, 2020 (Patients' medical history is fully available already before their baseline HbA1c test and was used to inform covariates)
Data exclusions	We pre-defined our retrospective research sample as adult patients, aged 18 to 80 years, registered with a GP practice and who received an HbA1c test between January 1st, 2017, and December 31st, 2018. We only included patients who did not exceed the HbA1c threshold for diabetes or prediabetes prior to their index date (i.e., date of their baseline HbA1c record) and who did not receive any diabetes medication prior to their index date (all codes available in our OSF repository) as these patients do not represent the target population of the NHS Diabetes Prevention Programme. We did not exclude any patients because of other criteria or metrics.
Non-participation	Due to the quasi-experimental/retrospective analysis strategy of routine patient data, non-participation is not an applicable category. It was not possible for patients to decline participation for this specific manuscript. However, we report missingness in outcomes (i.e., what is generally understood as loss to follow-up in routine data) in Supplementary Table S1.
Randomization	Randomization is not actively performed in the quasi-experimental analysis strategies that we used in this manuscript (regression discontinuity analysis, difference-in-differences analysis, instrumental variable estimation). In each of the analysis strategies, confounding is addressed distinctively. For the regression discontinuity analysis, the arbitrary nature of the prediabetes HbA1c threshold along with the fact that HbA1c is measured with a degree of random measurement error means that patients just below and above the threshold should be close to identical in their underlying characteristics. We demonstrated that there is no sudden "jump" at the prediabetes HbA1c threshold in age, baseline BMI, and frequency of GP visits, signifying that these variables cannot confound our analysis. We demonstrate this further using balance tests, empirically showing that a comprehensive set of potential confounding variables, including demographic characteristics, physiological variables, and markers of healthcare utilization and literacy, were all continuously distributed near the threshold (figs. S2-3, Extended Data Table 2). Taken together, our analyses provide strong evidence that the necessary assumptions for a valid regression discontinuity analysis were met, meaning that patients below and above the threshold were effectively randomized to being referred to the program or not as they would be in an ideal controlled trial (Supplementary Information 1.3). For the difference-in-differences analysis, a key assumption is that, in the absence of treatment, the outcome trends would have been the same in the treatment group as in the comparison group. We interrogated this assumption by examining whether the trends in HbA1c are visually parallel in the pre-introduction period (Extended Data Fig. 1) and by testing the interaction between wave and (pre-introduction) year in linear regressions (Wald test: Wave 1, $p = 0.273$, wave 2, $p = 0.809$). For the Instrumental variable (IV) estimation, we addressed endogeneity in regression analysis (that means the fact that being referred to the NHS DPP is correlated with the error term, e.g. due to selection bias in treatment assignment) by using regional variation in NHS DPP coverage as an exogenous instrument for the patient-level probability of being referred to the NHS DPP.

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n/a	Involvement in the study
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☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involvement in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging